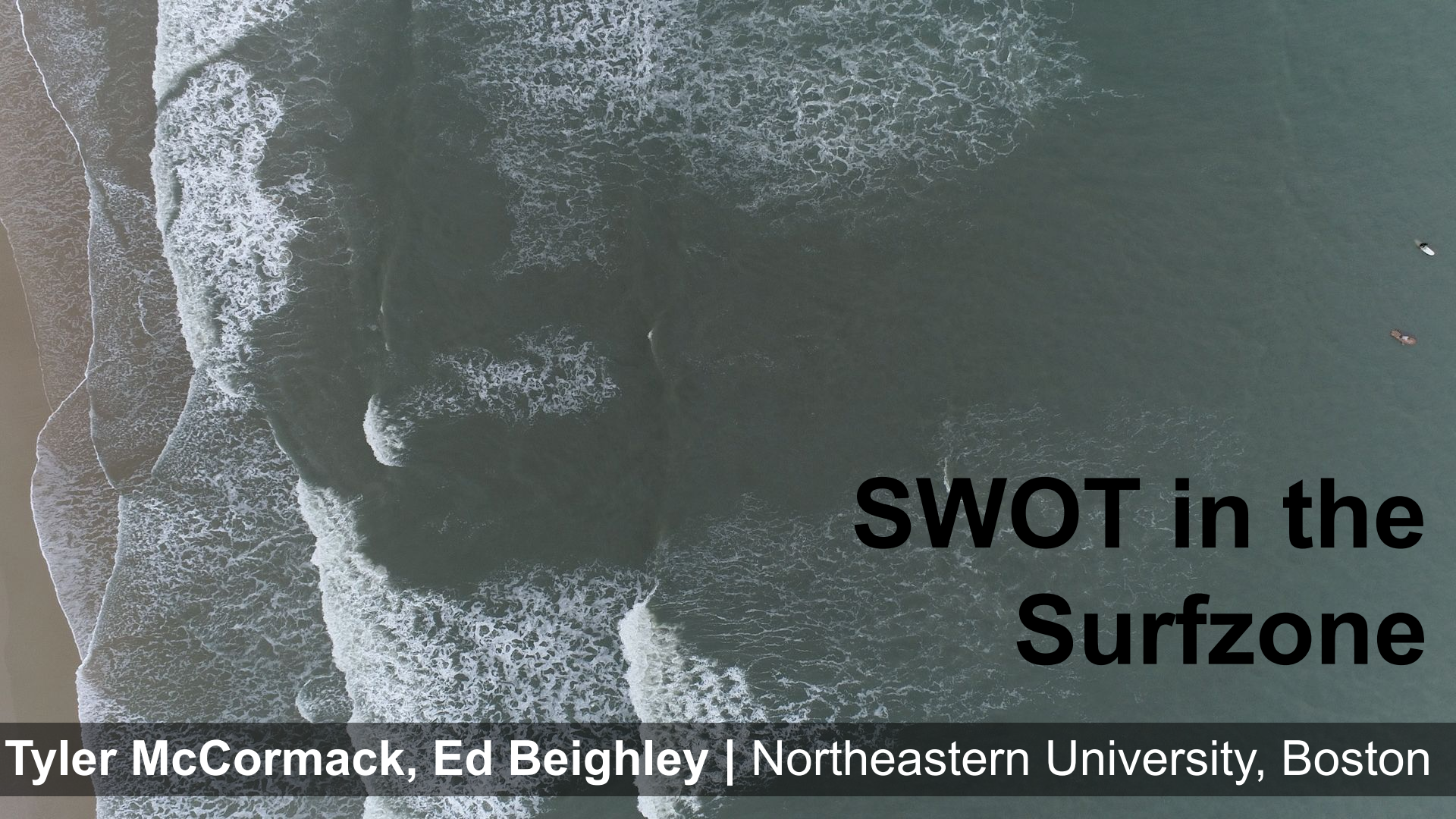




SWOT in the Surfzone

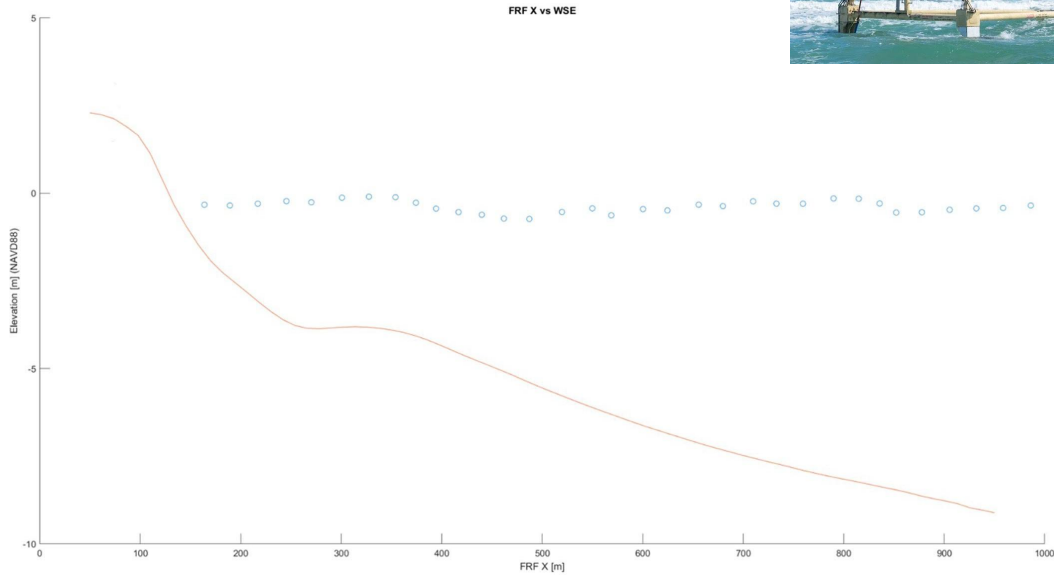
Tyler McCormack, Ed Beighley | Northeastern University, Boston



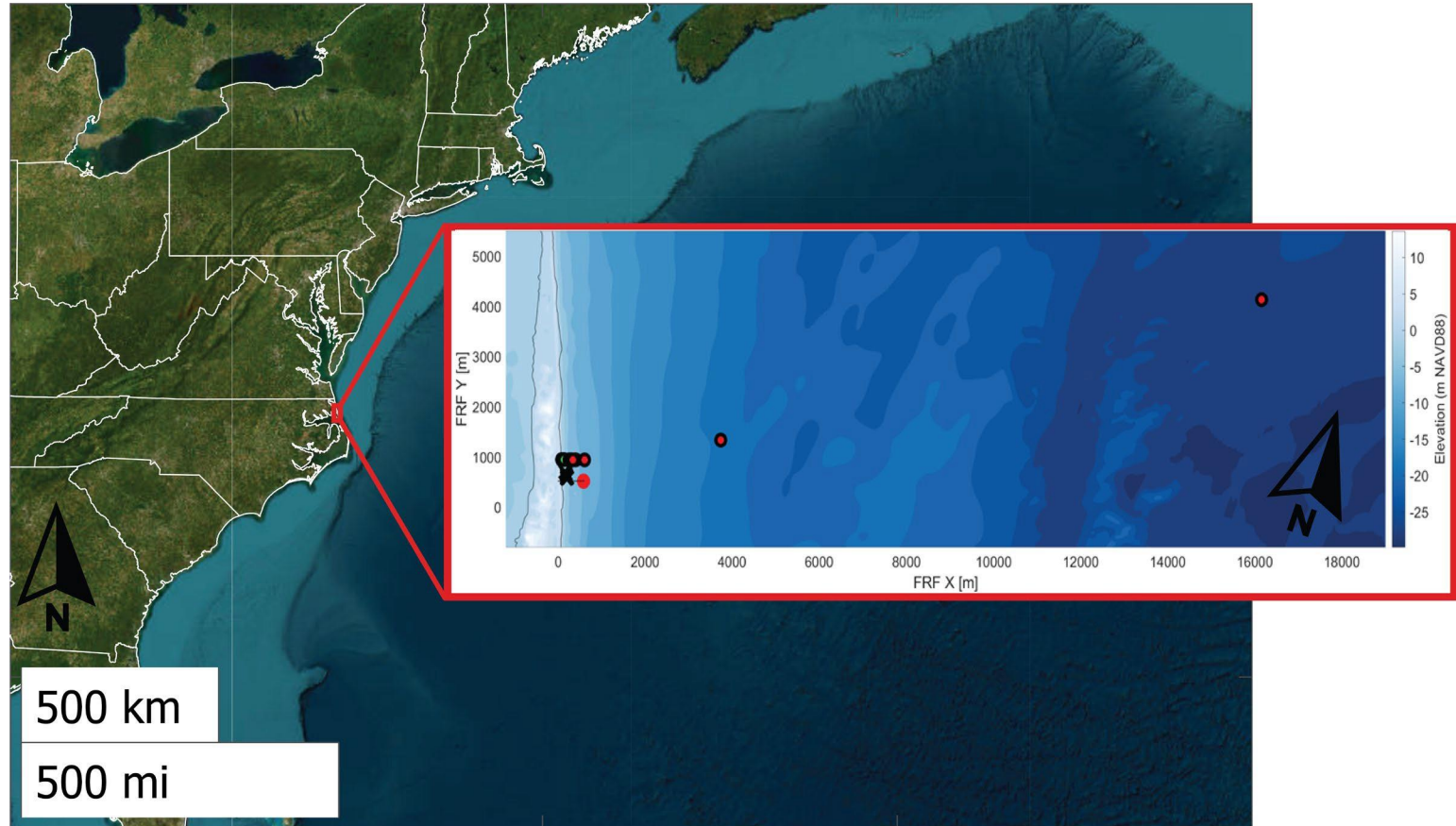
Surfzone

US Army Corps of Engineers Field Research Facility, Duck NC

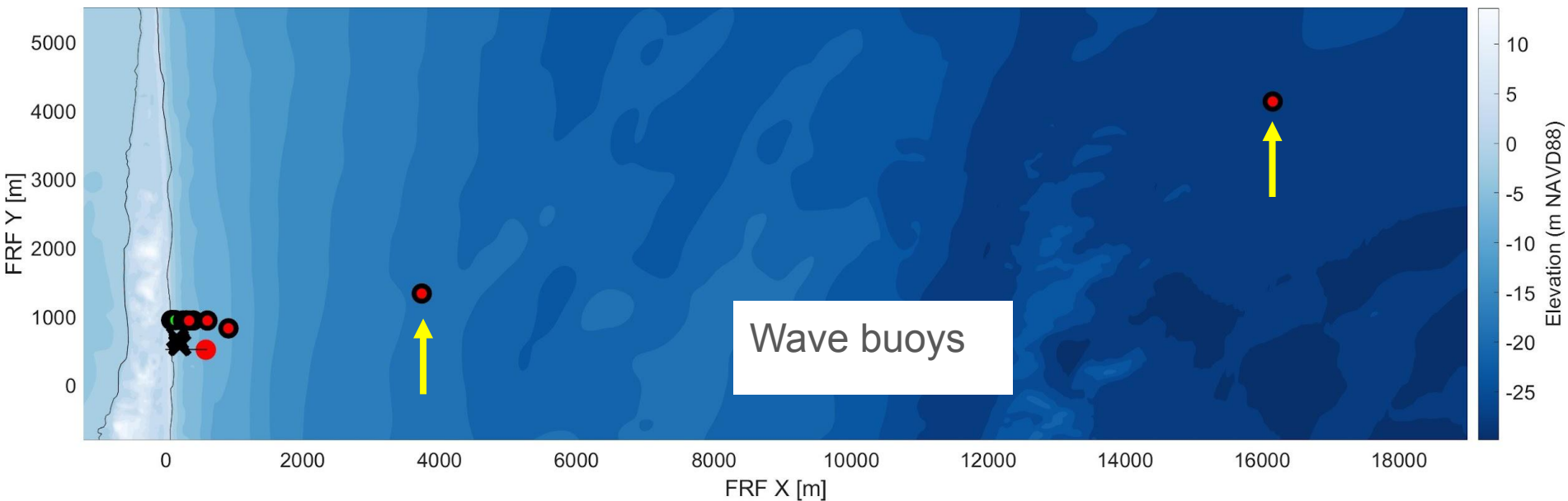
- Continuous water level and wave gage measurements to compare SWOT data to
- Regular bathymetric surveys



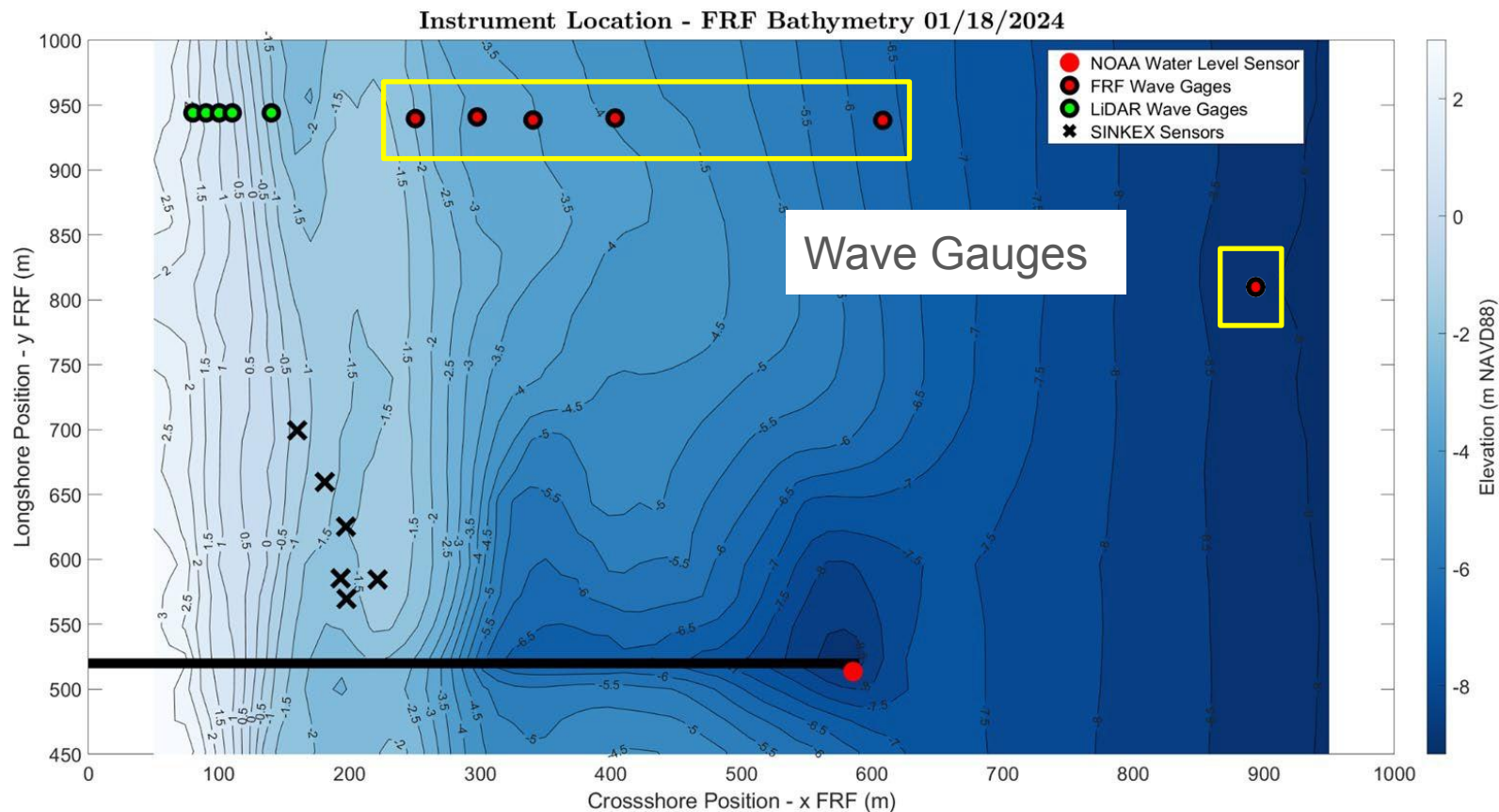
US Army Corp Field Research Facility, Duck NC



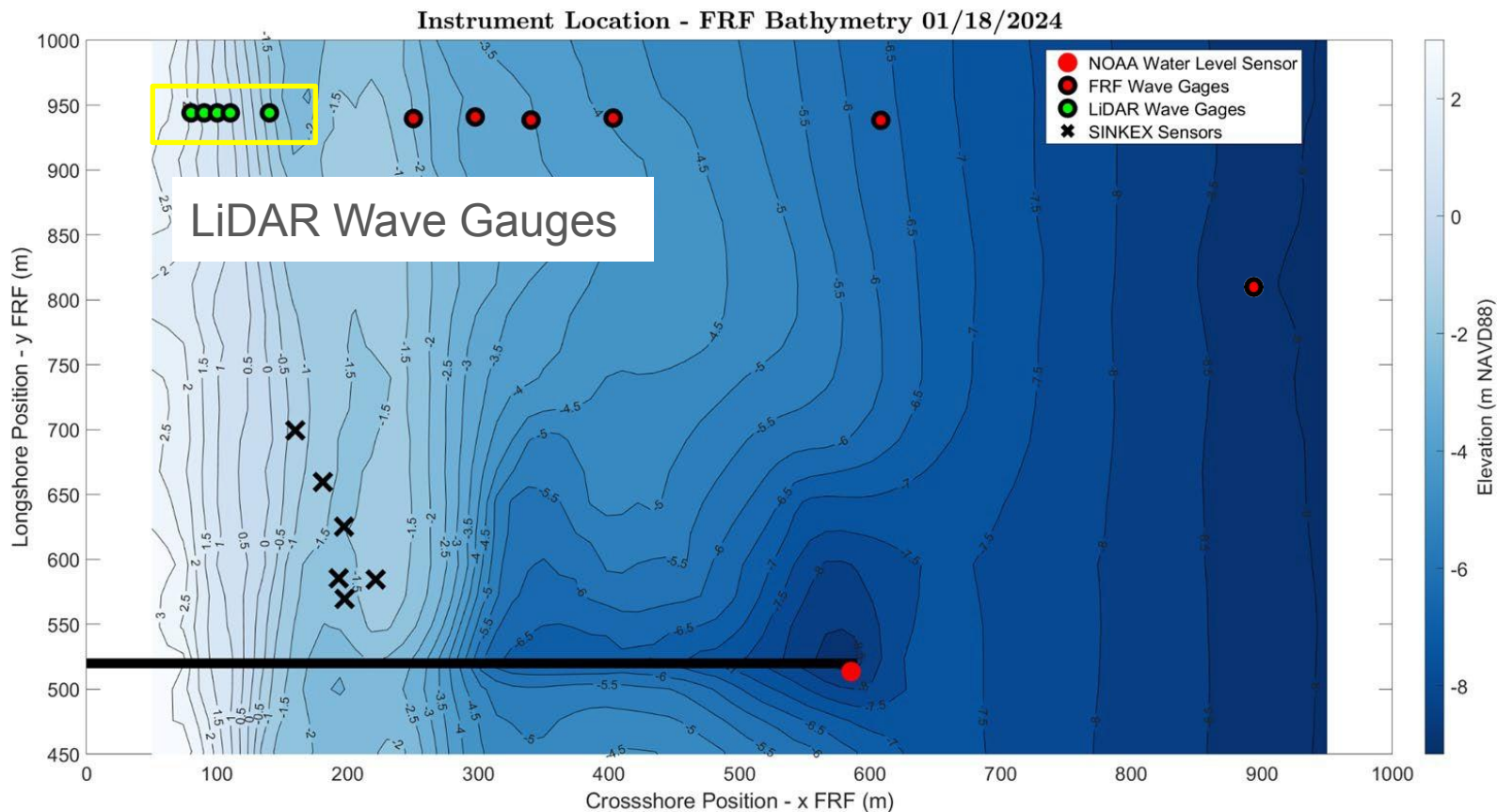
US Army Corp Field Research Facility, Duck NC



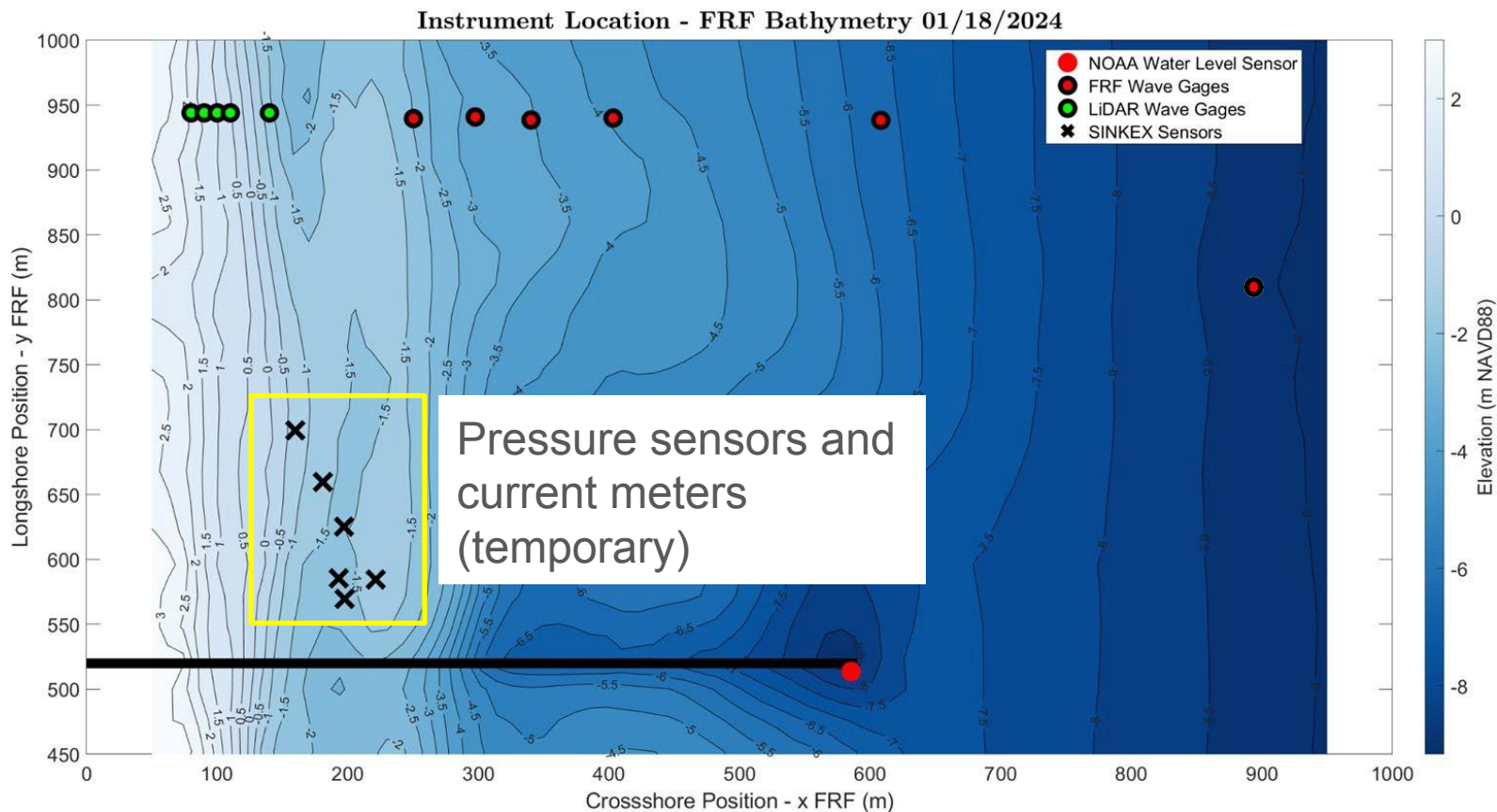
US Army Corp Field Research Facility, Duck NC



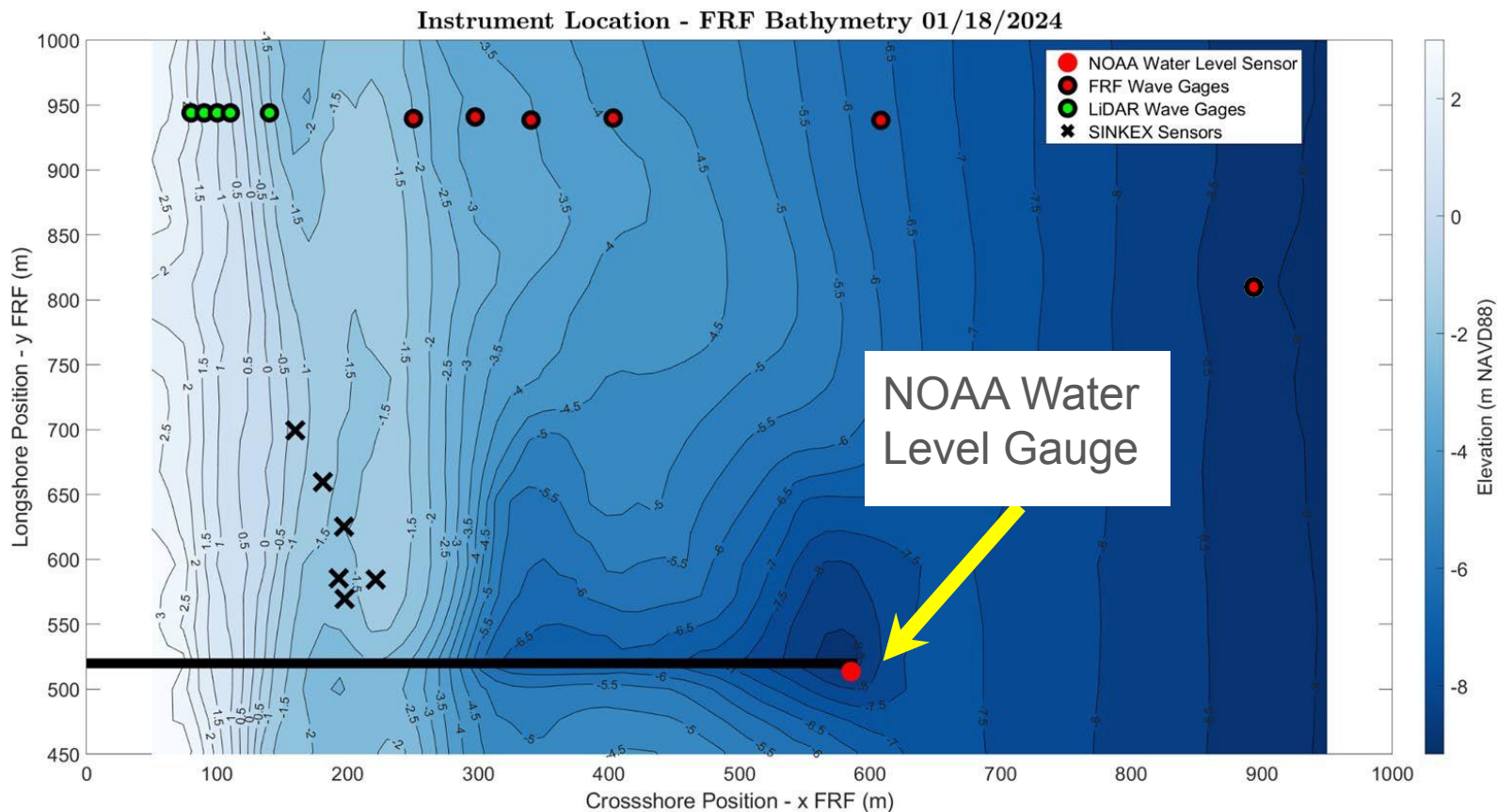
US Army Corp Field Research Facility, Duck NC



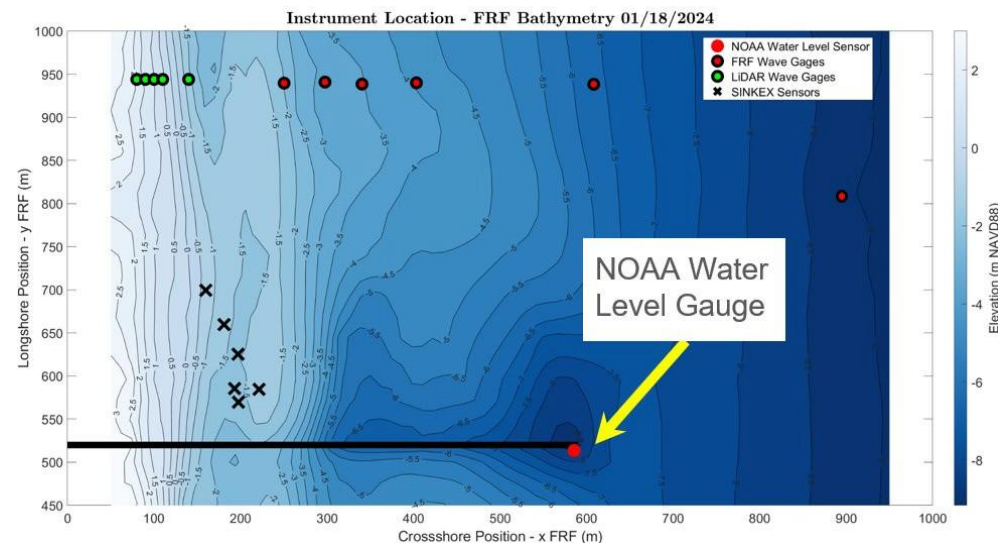
US Army Corp Field Research Facility, Duck NC



US Army Corp Field Research Facility, Duck NC

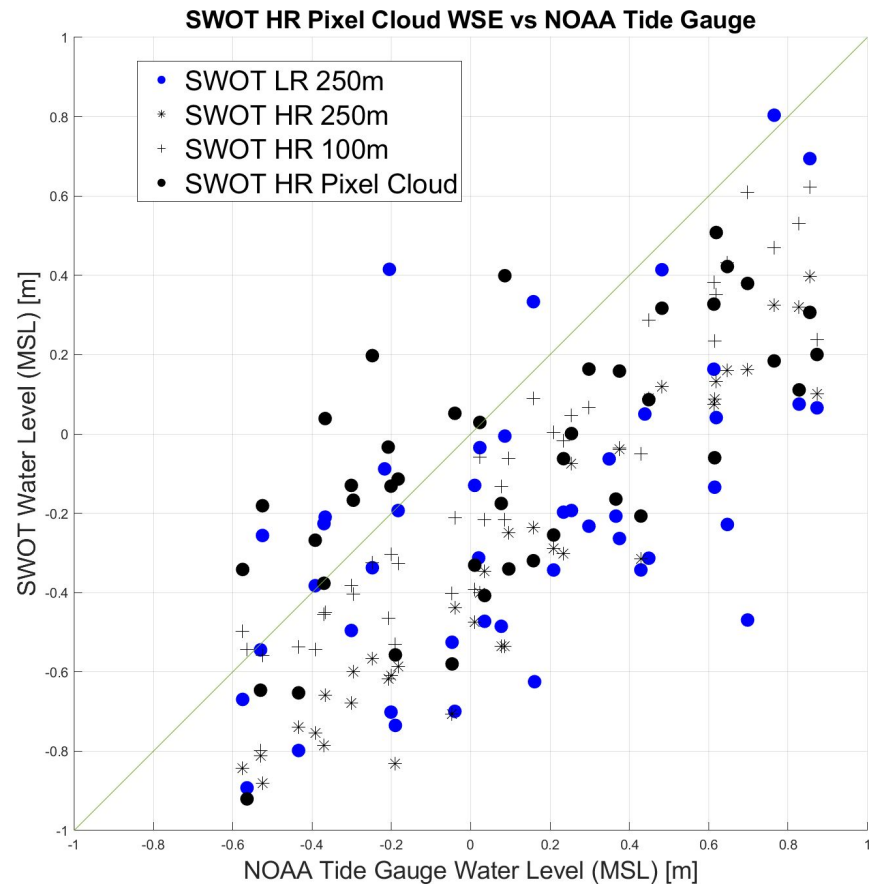


Pixel Cloud WSE Accuracy in the Surfzone



Error Metrics

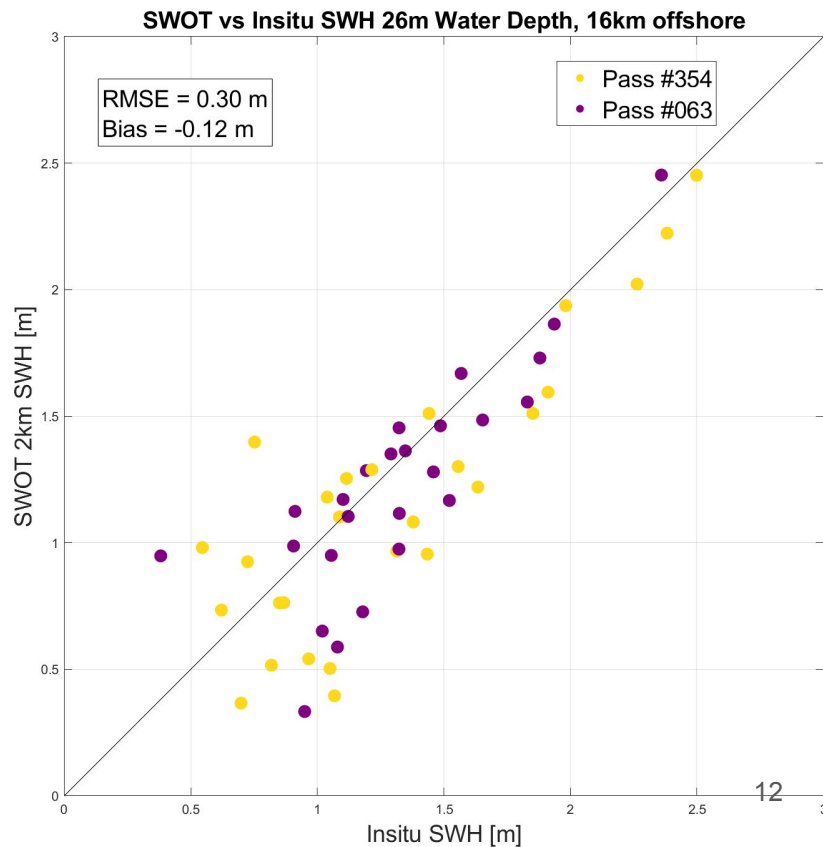
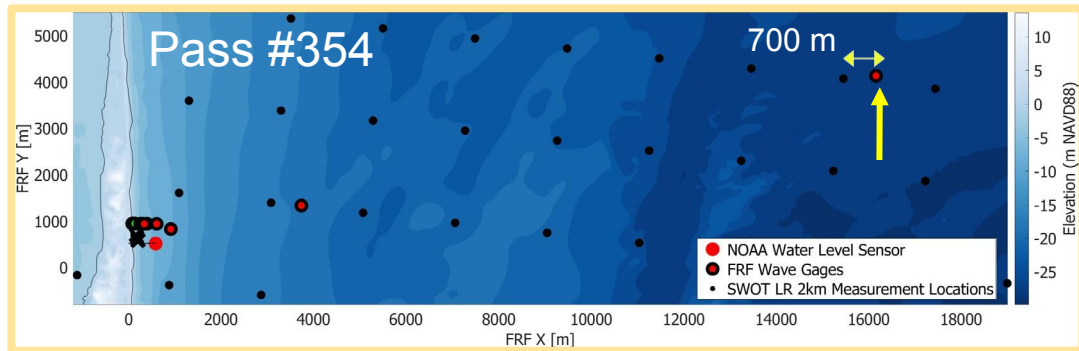
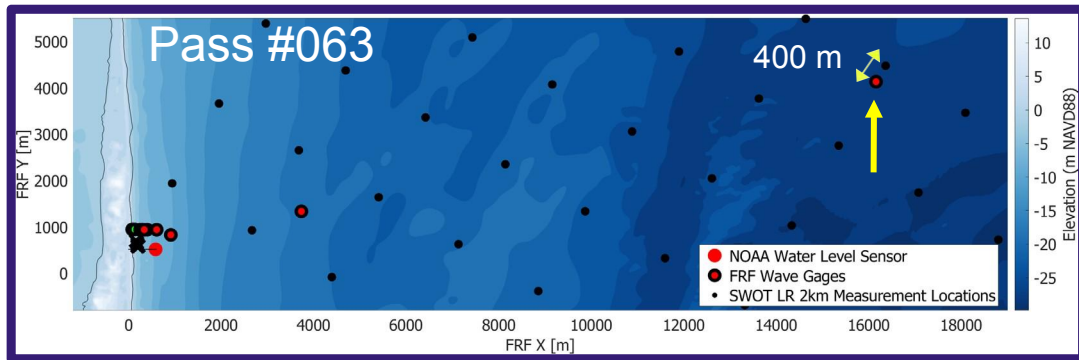
SWOT Product	RMSE [m]	Bias [m]	R^2
● LR 250m	0.56	-0.39	0.30
* HR 250m	0.46	-0.45	0.93
+ HR 100m	0.25	-0.21	0.91
● HR Pixel Cloud	0.37	-0.19	0.47



Native SWOT Significant Wave Height (SWH) Accuracy

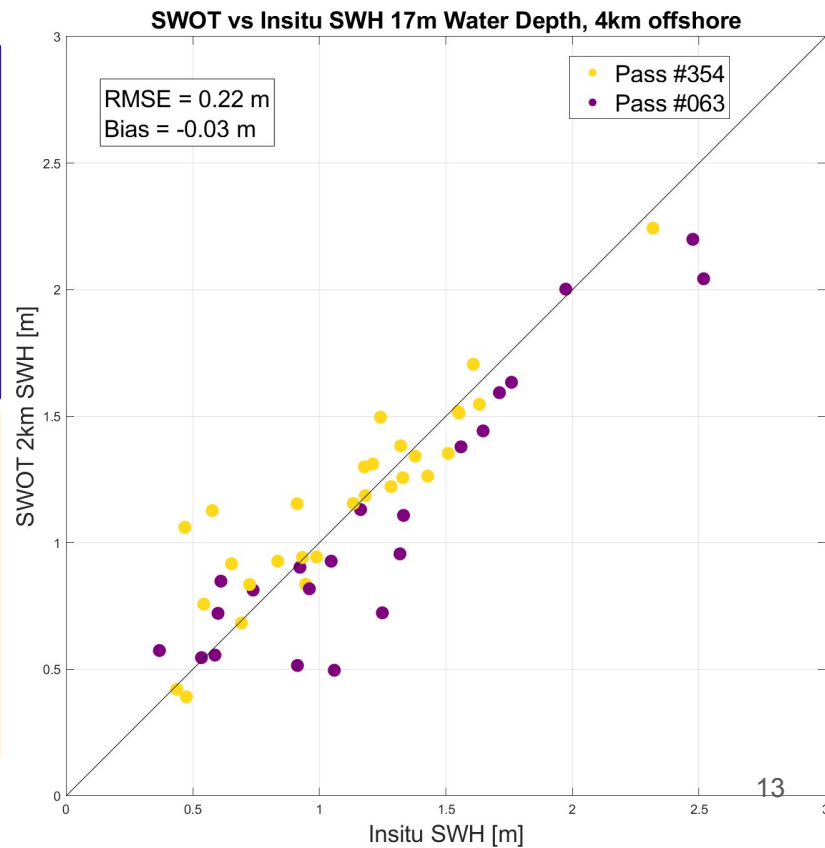
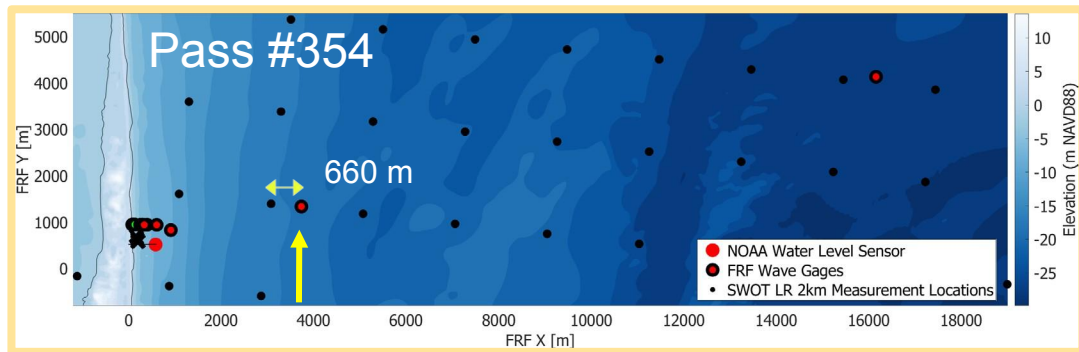
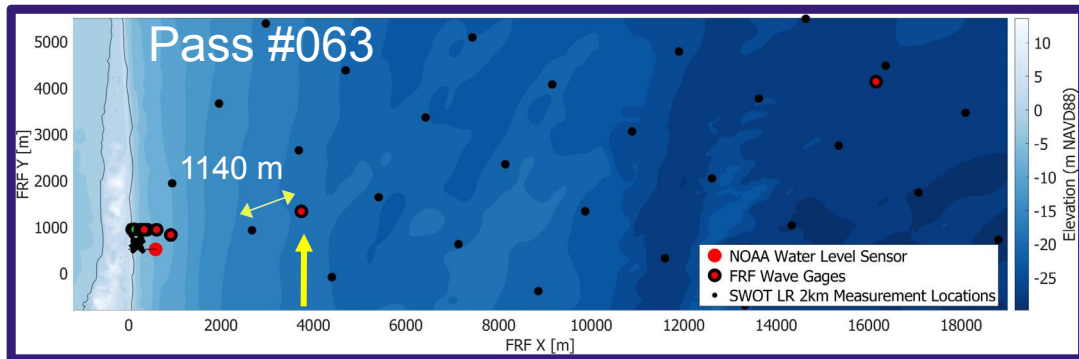
Native SWOT Significant Wave Height (SWH) Accuracy

Wave Buoy, 26m water depth, 16 km offshore



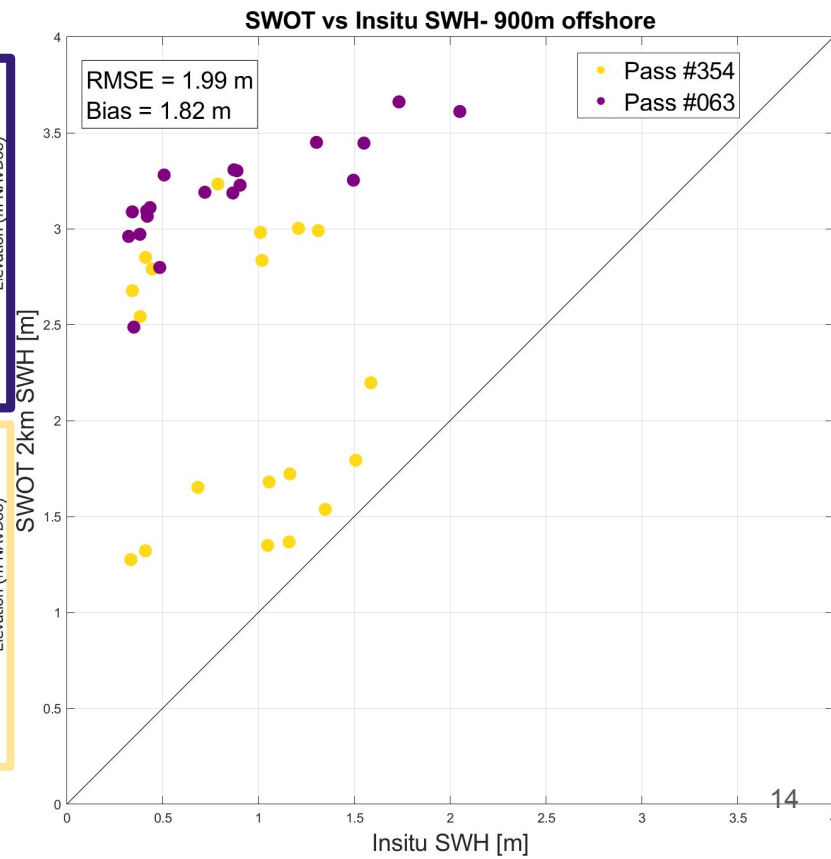
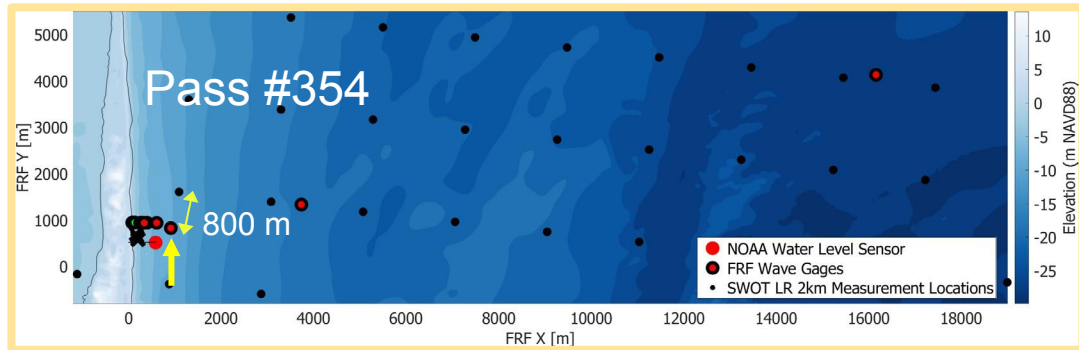
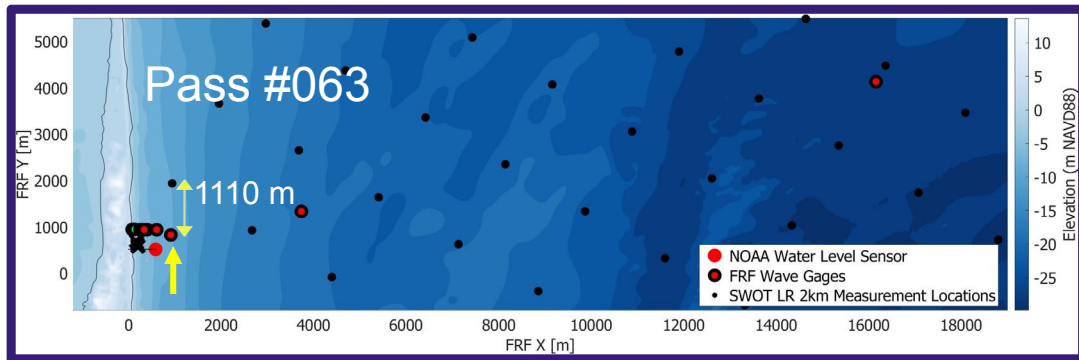
Native SWOT Significant Wave Height (SWH) Accuracy

Wave Buoy, 17m water depth, 4 km offshore



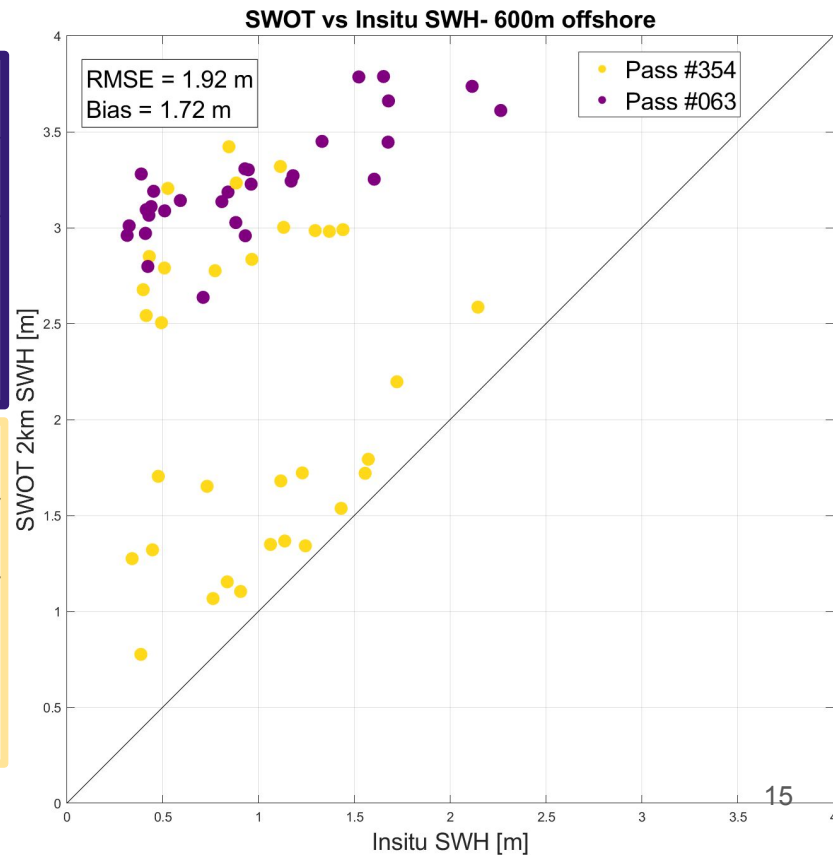
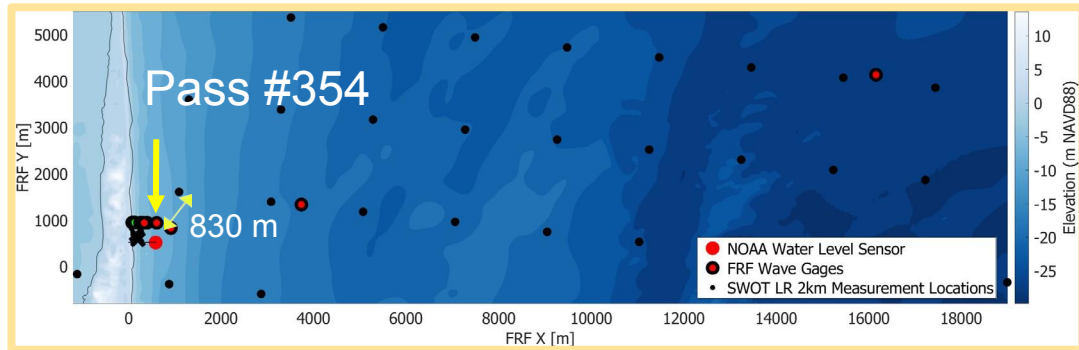
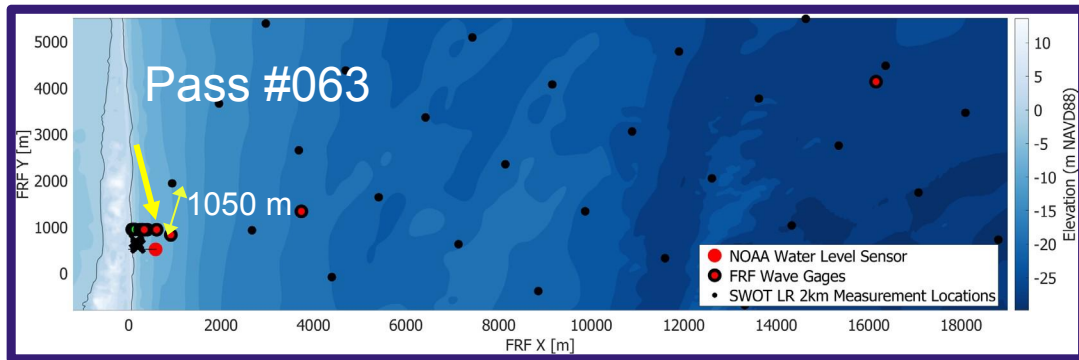
Native SWOT Significant Wave Height (SWH) Accuracy

Pressure sensor, 8m water depth, 900m offshore

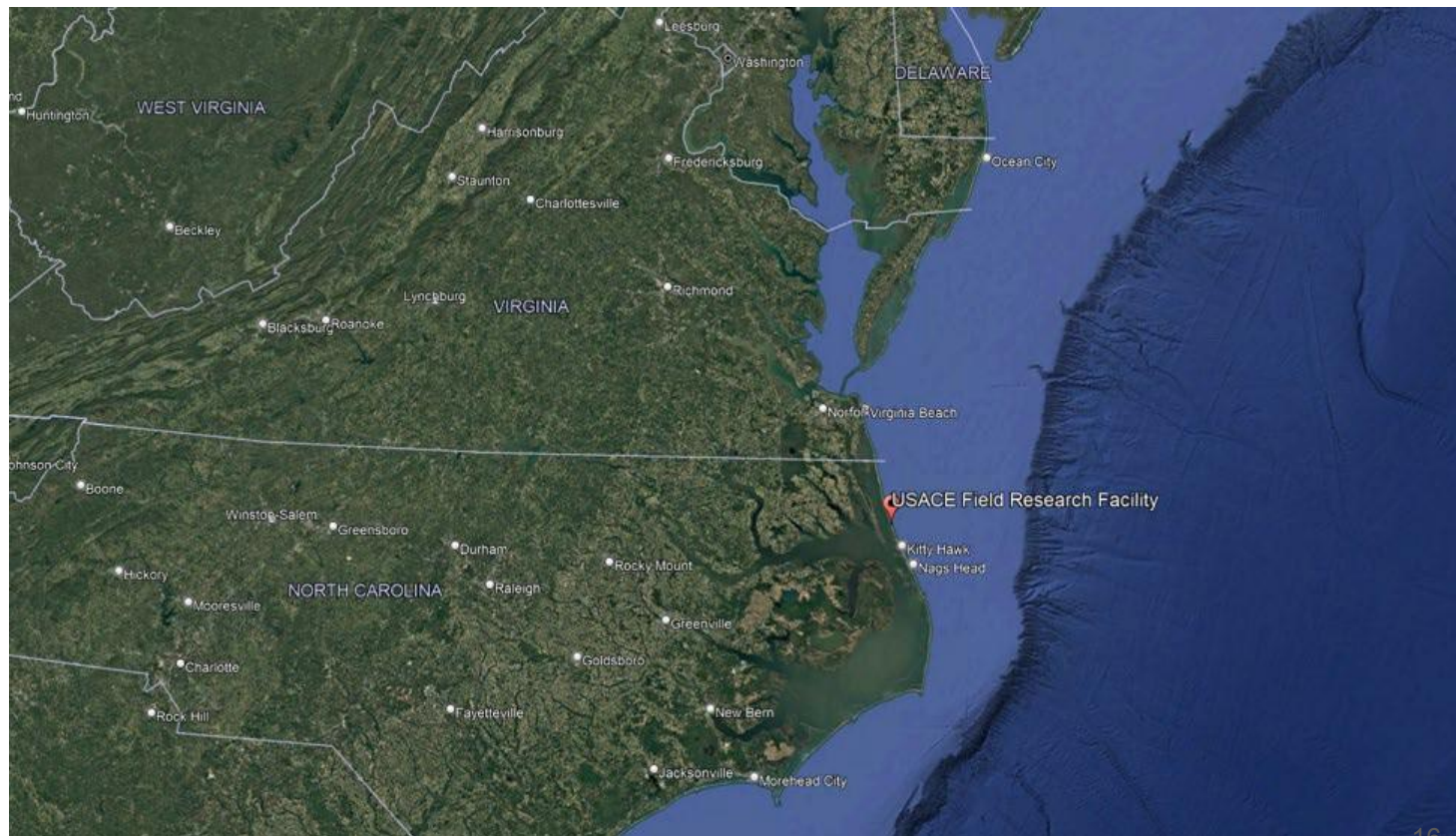


Native SWOT Significant Wave Height (SWH) Accuracy

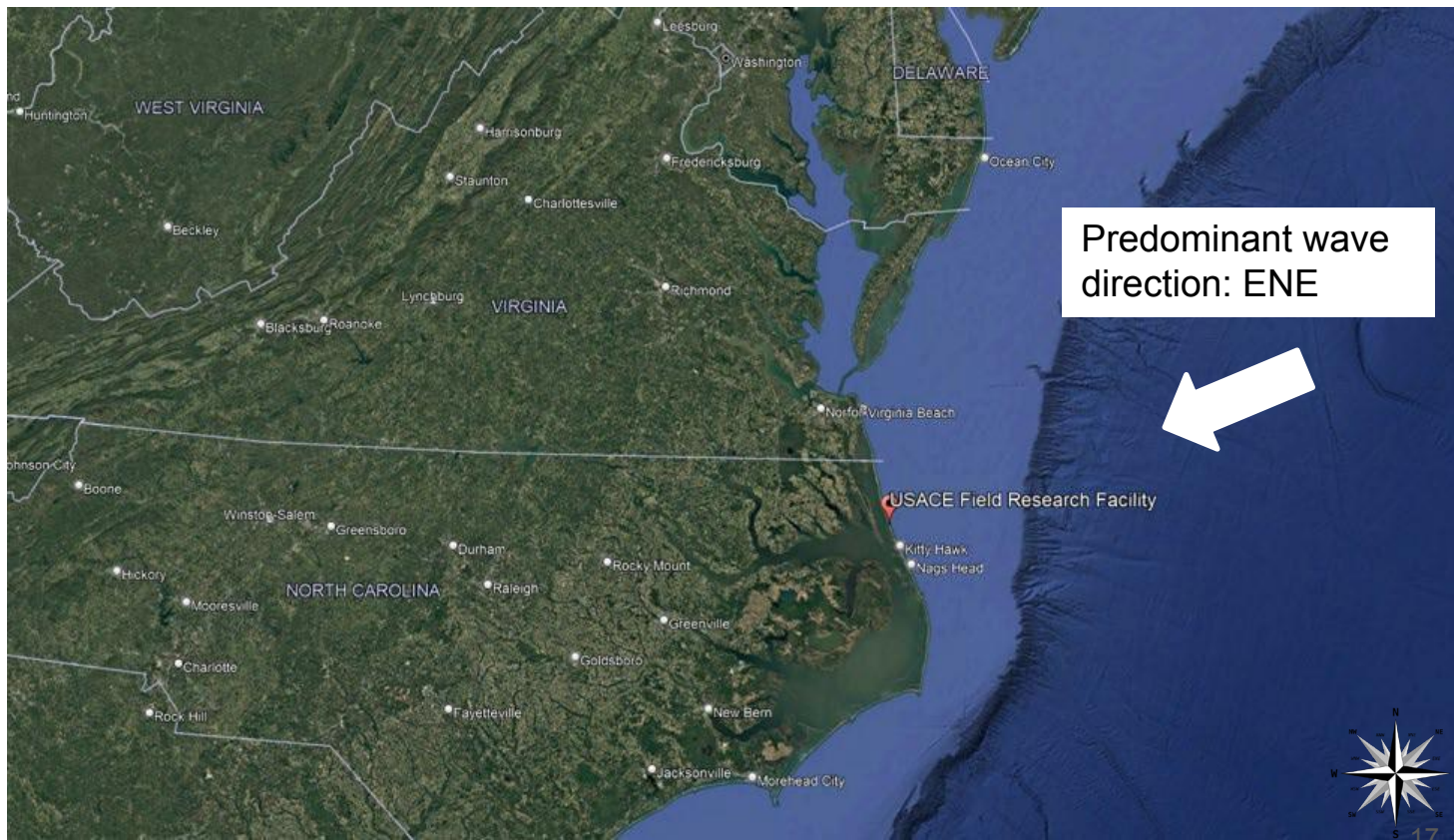
Pressure sensor, 6m water depth, 600m offshore



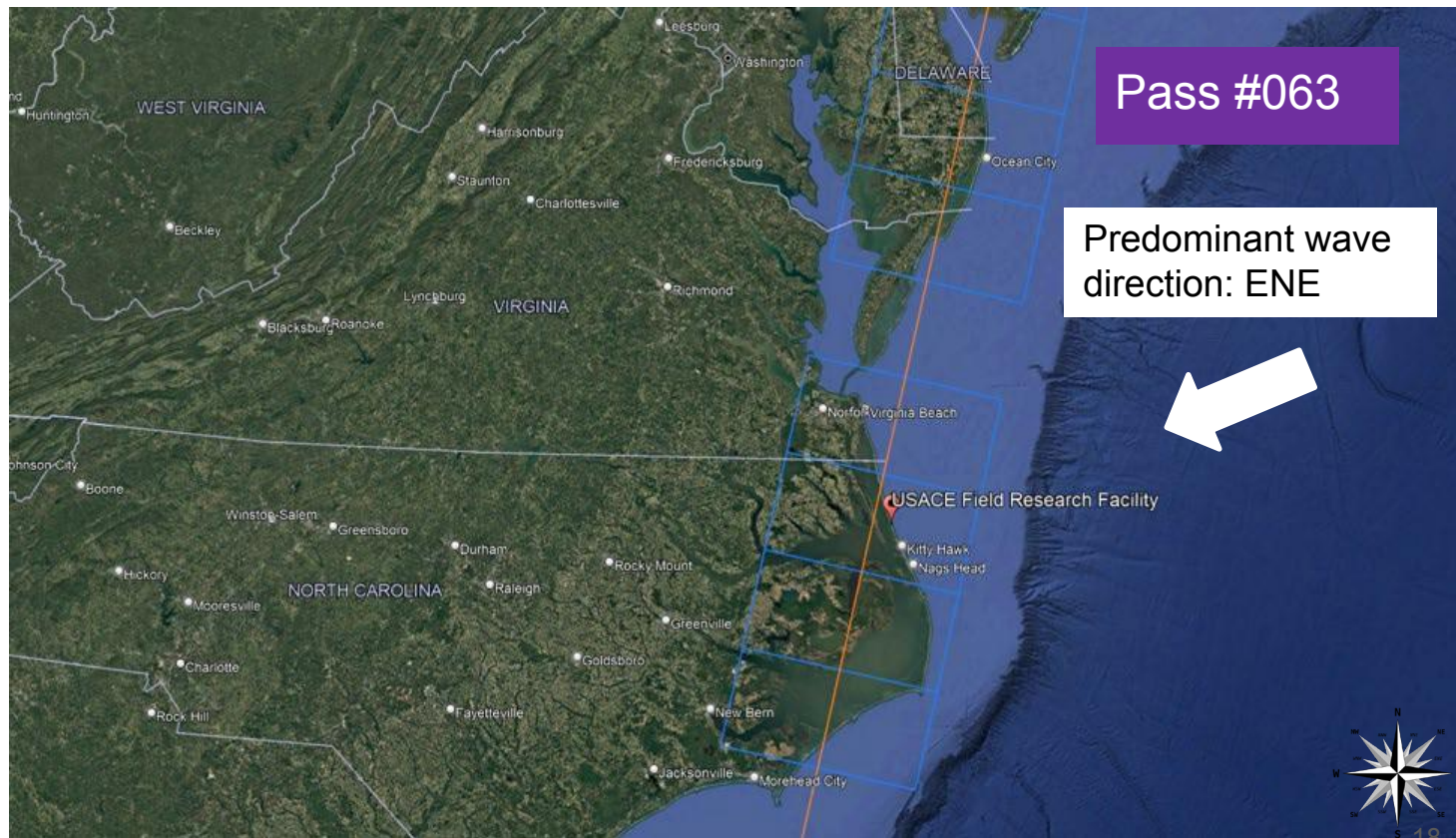
Effect of Pass Orientation on SWH Accuracy



Effect of Pass Orientation on SWH Accuracy



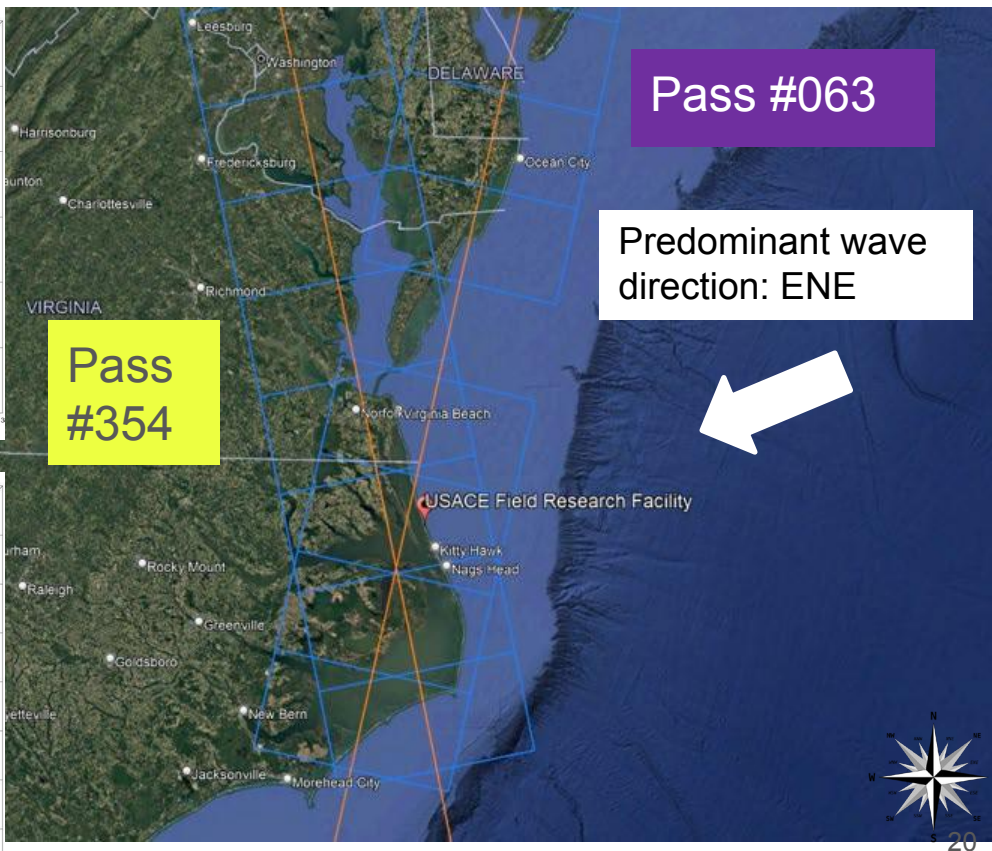
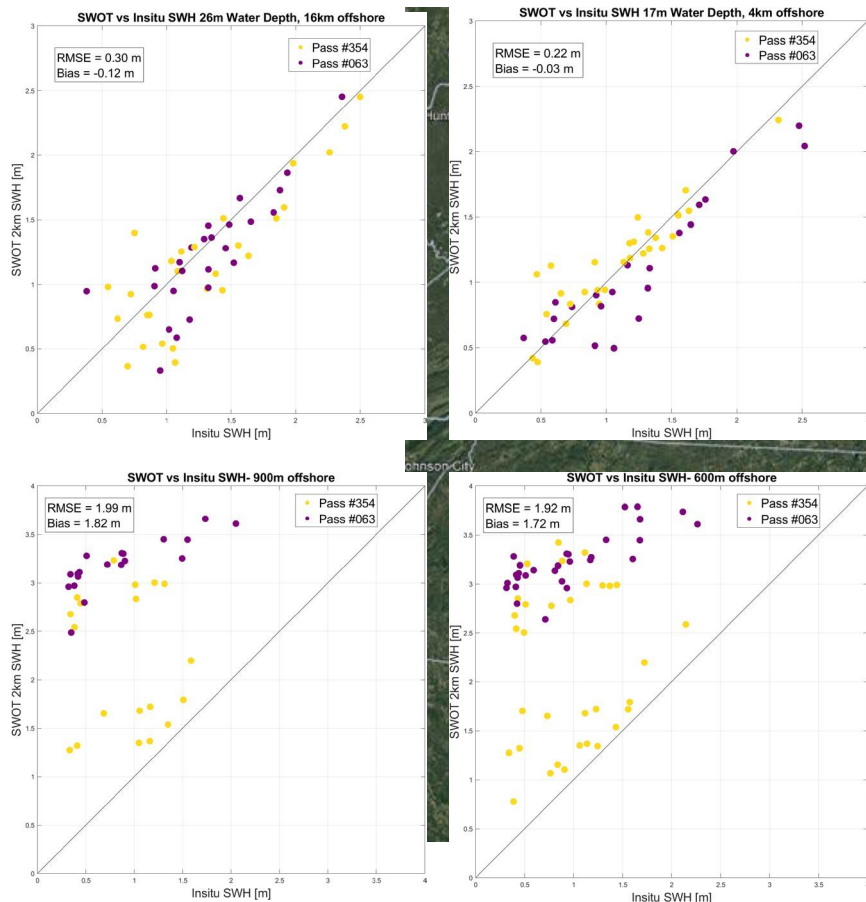
Effect of Pass Orientation on SWH Accuracy



Effect of Pass Orientation on SWH Accuracy

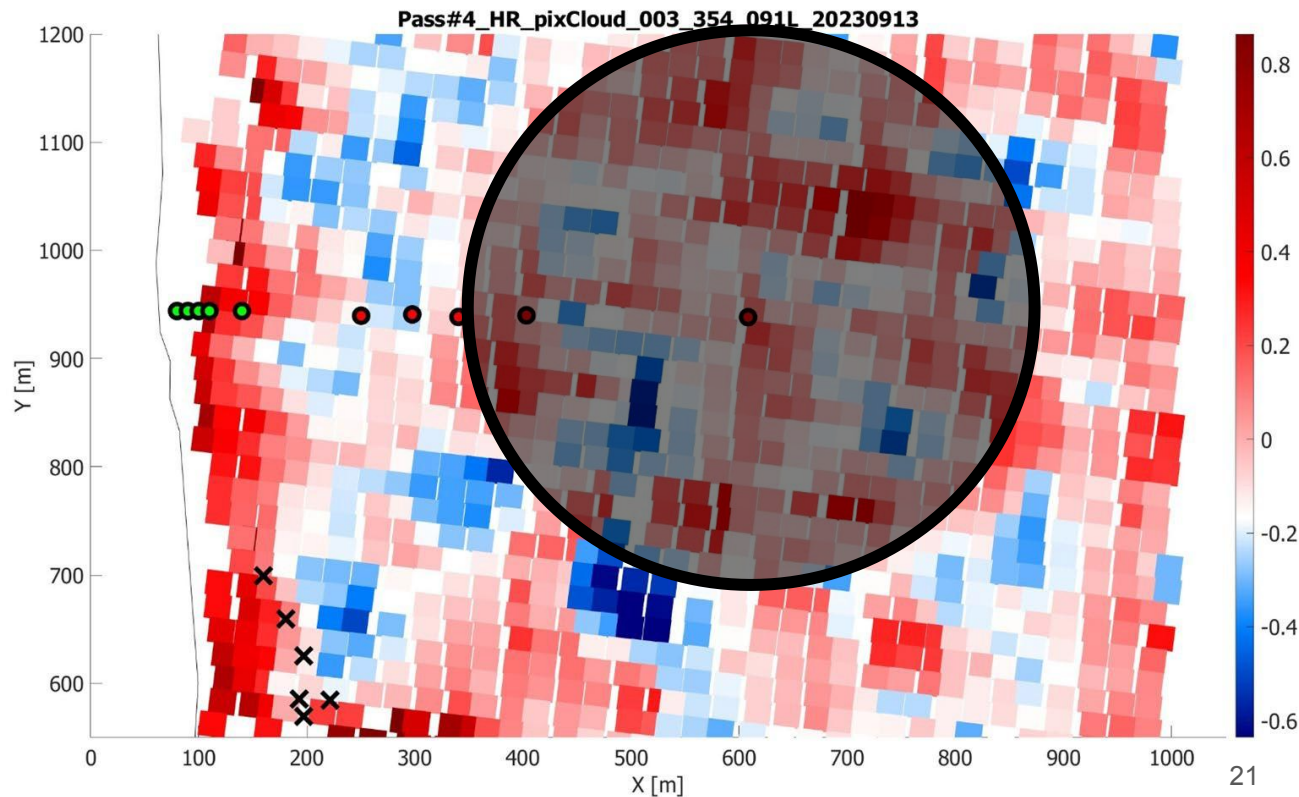


Effect of Pass Orientation on SWH Accuracy

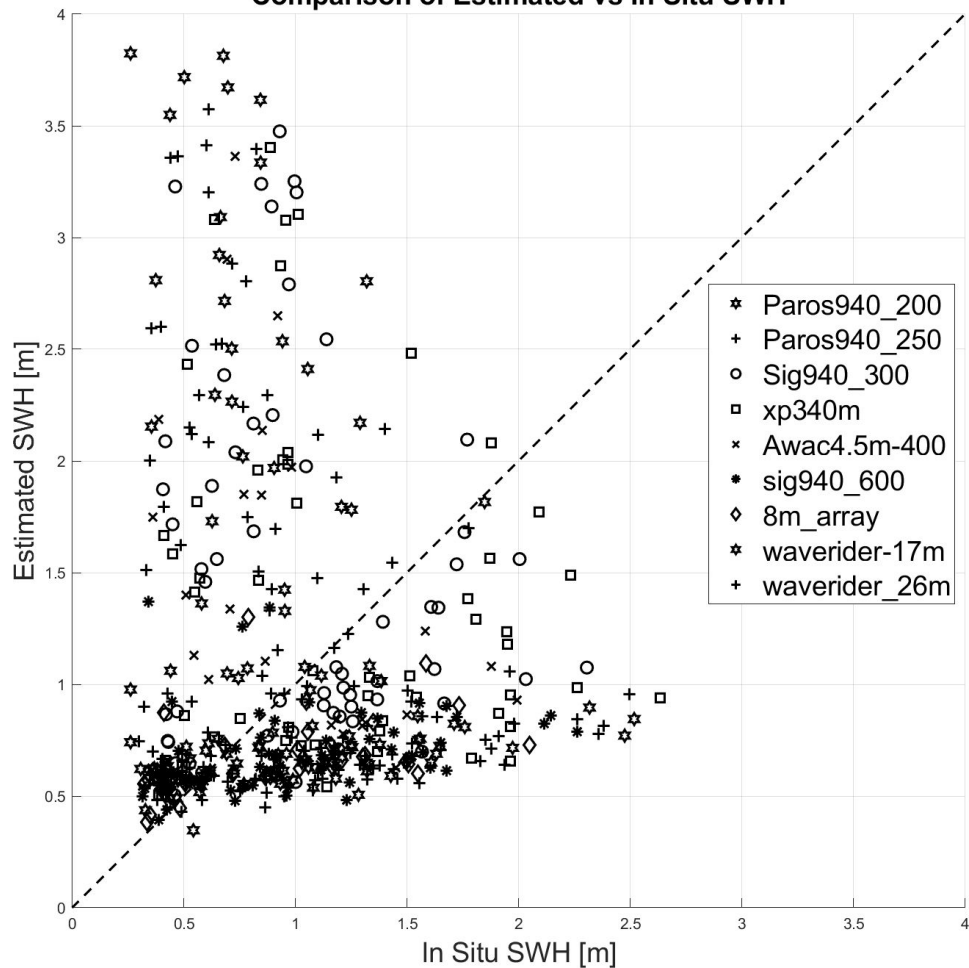


Pixel Cloud WSE based SWH measurements

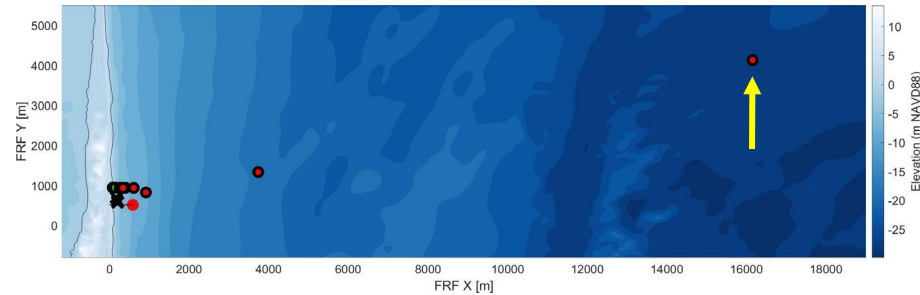
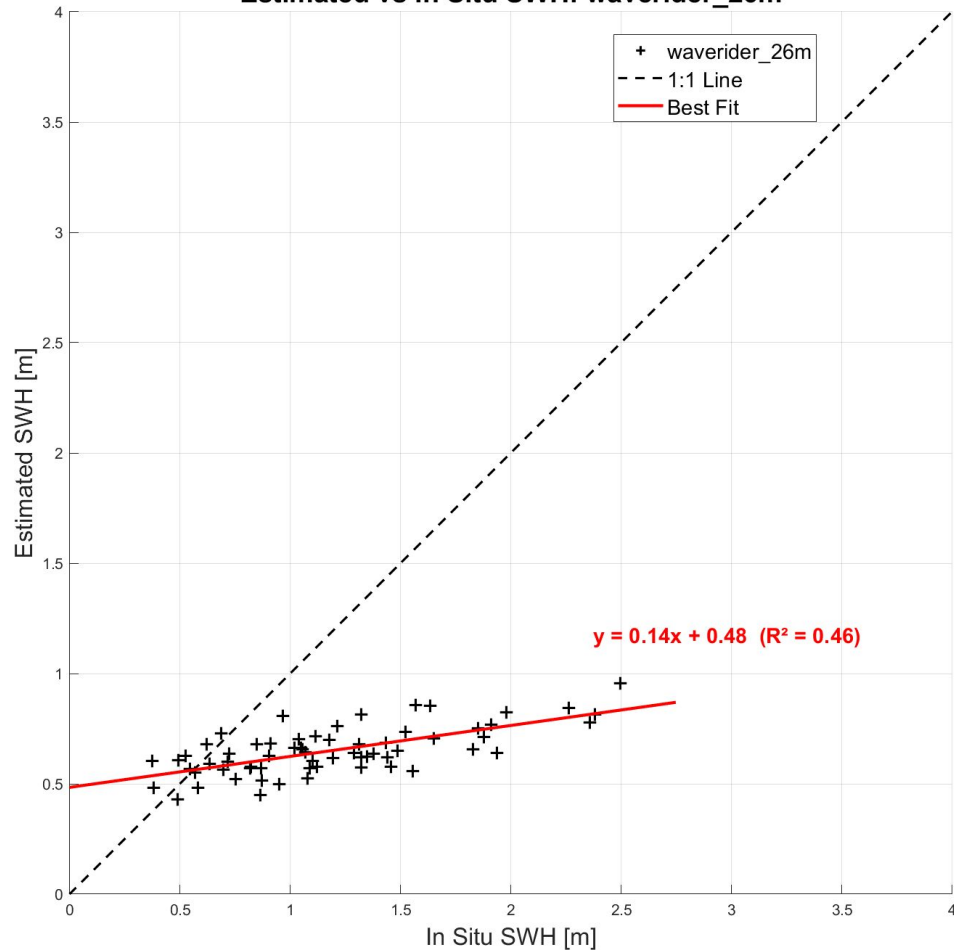
$$\text{SWH} = 4 * \text{STD}(\text{WSE})$$



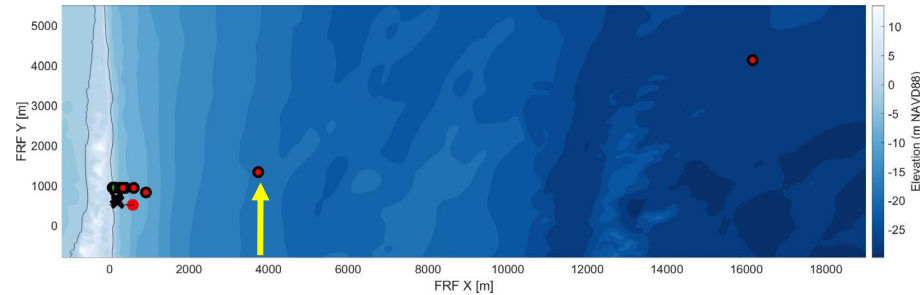
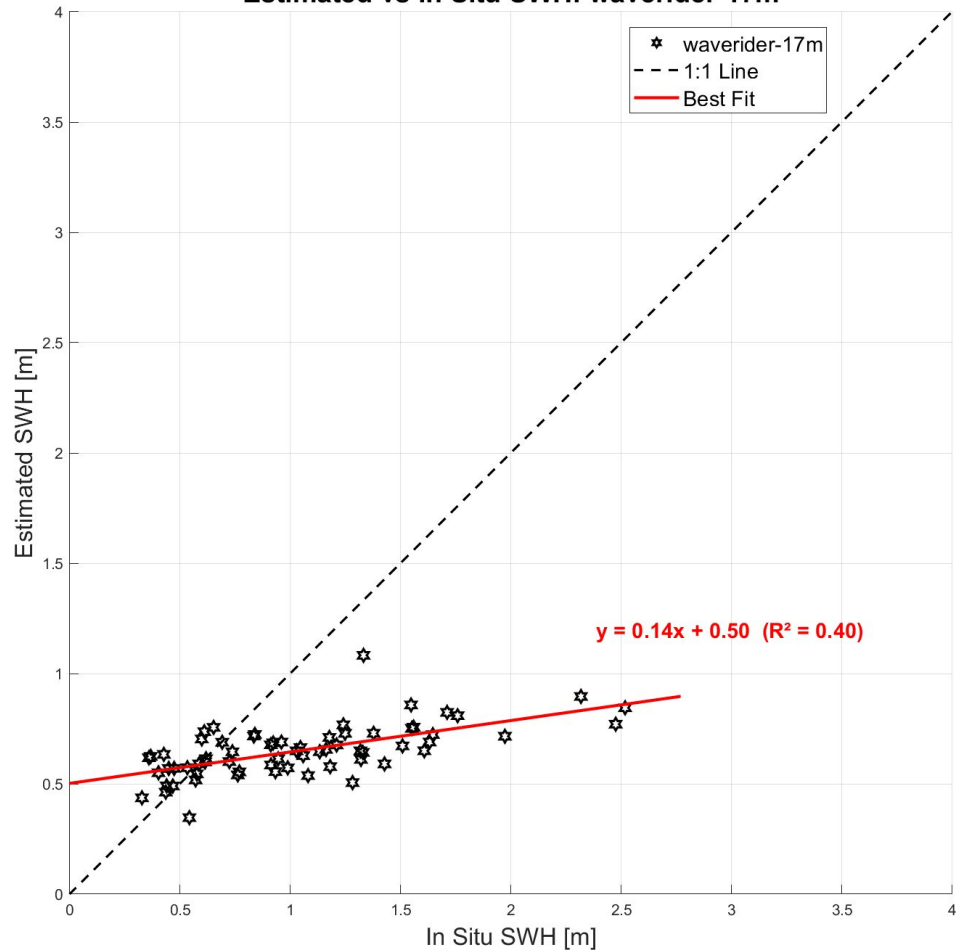
Comparison of Estimated vs In Situ SWH



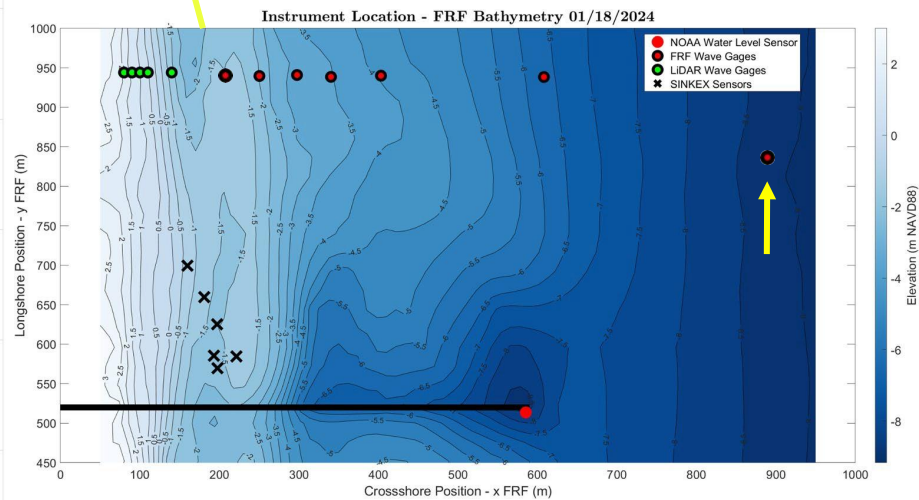
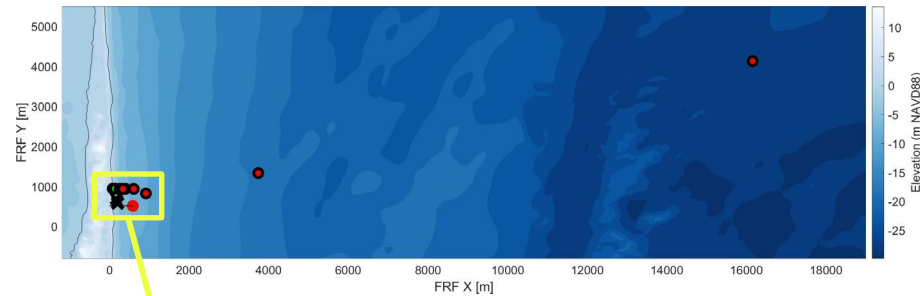
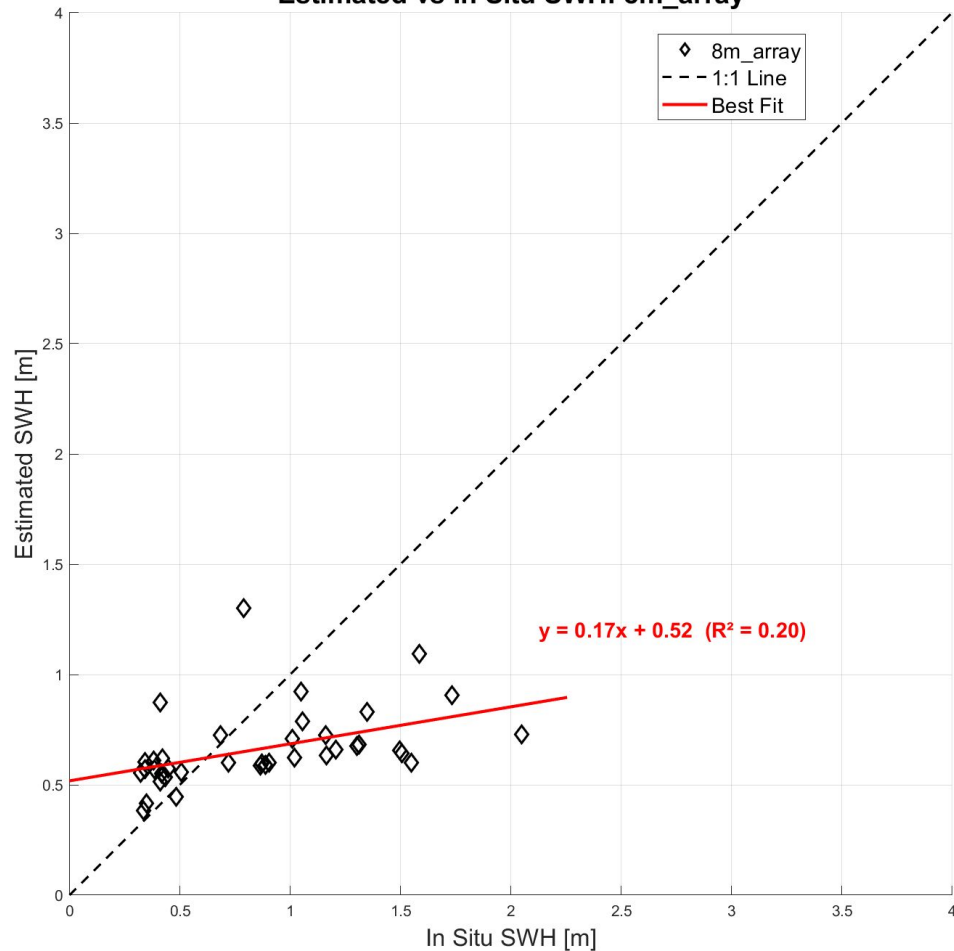
Estimated vs In Situ SWH: waverider_26m



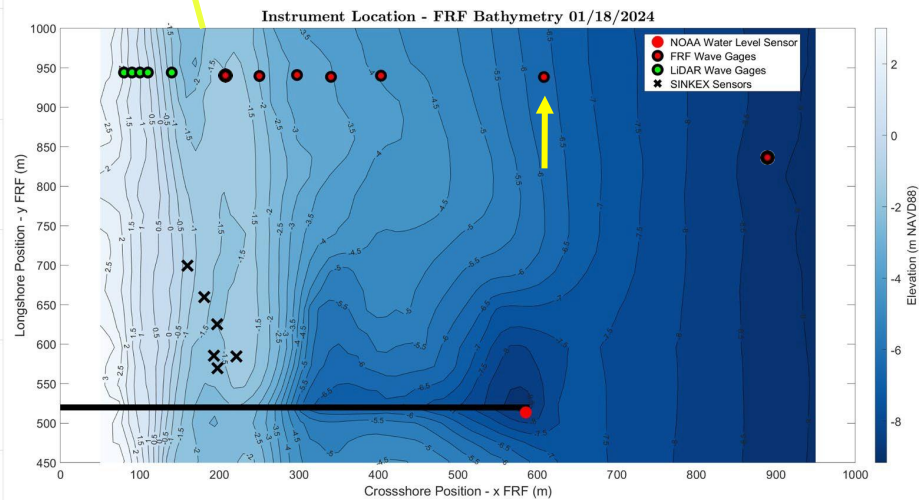
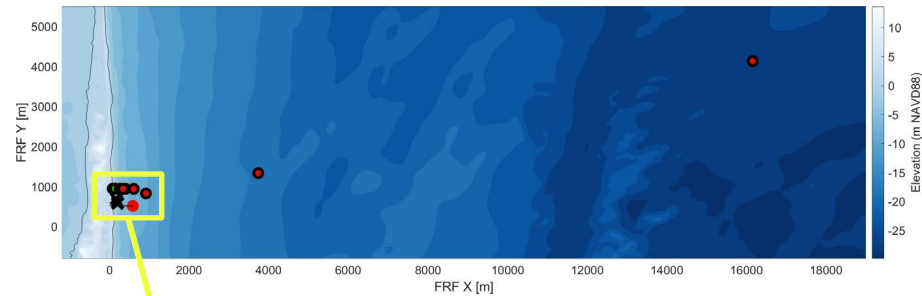
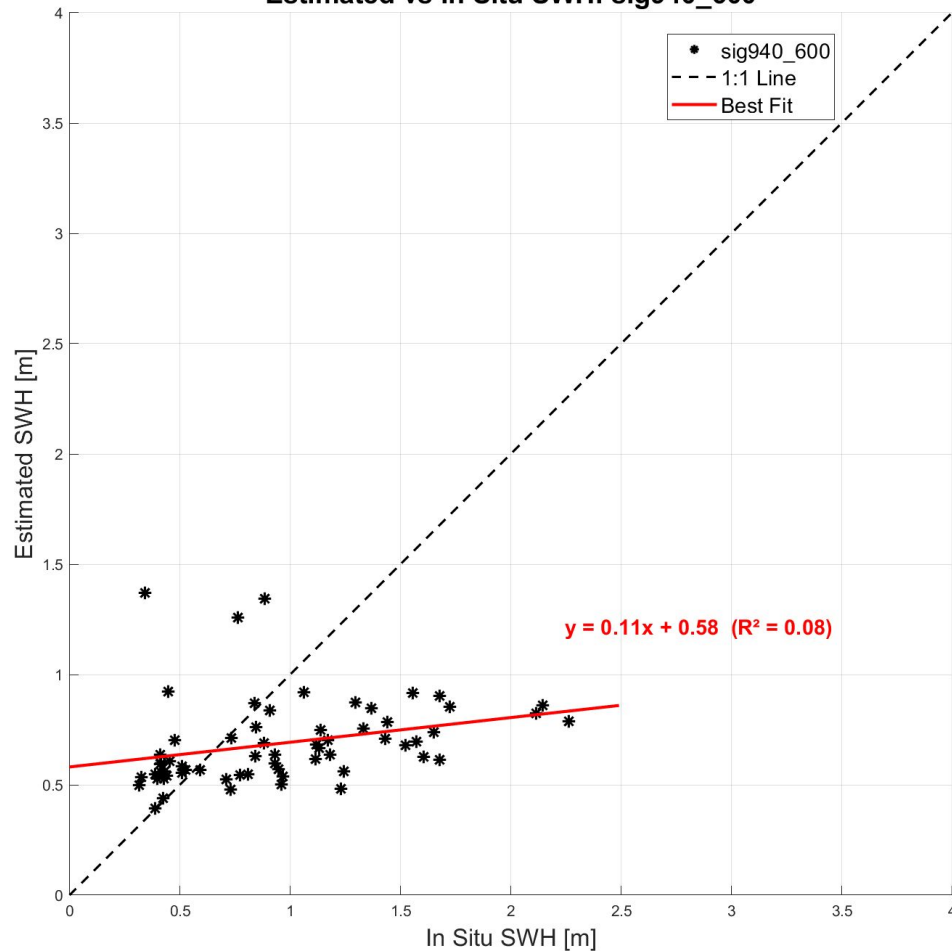
Estimated vs In Situ SWH: waverider-17m



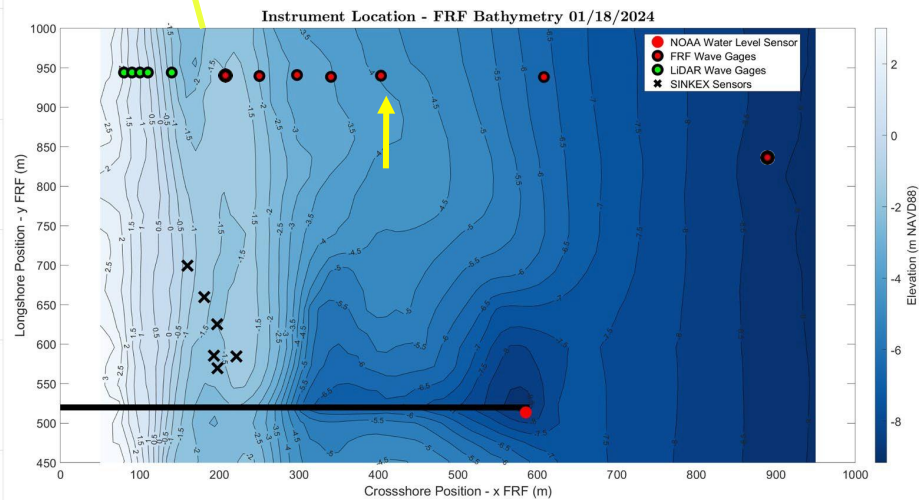
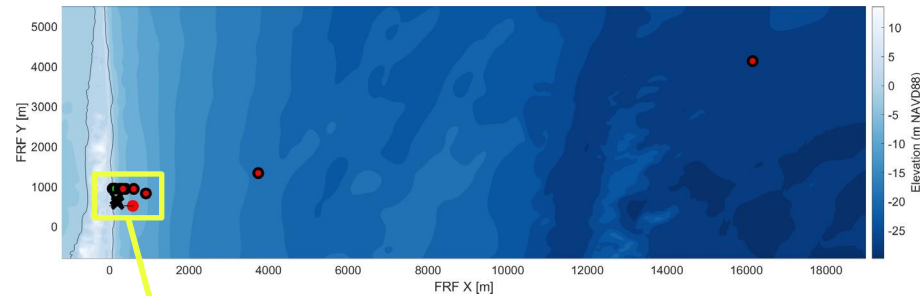
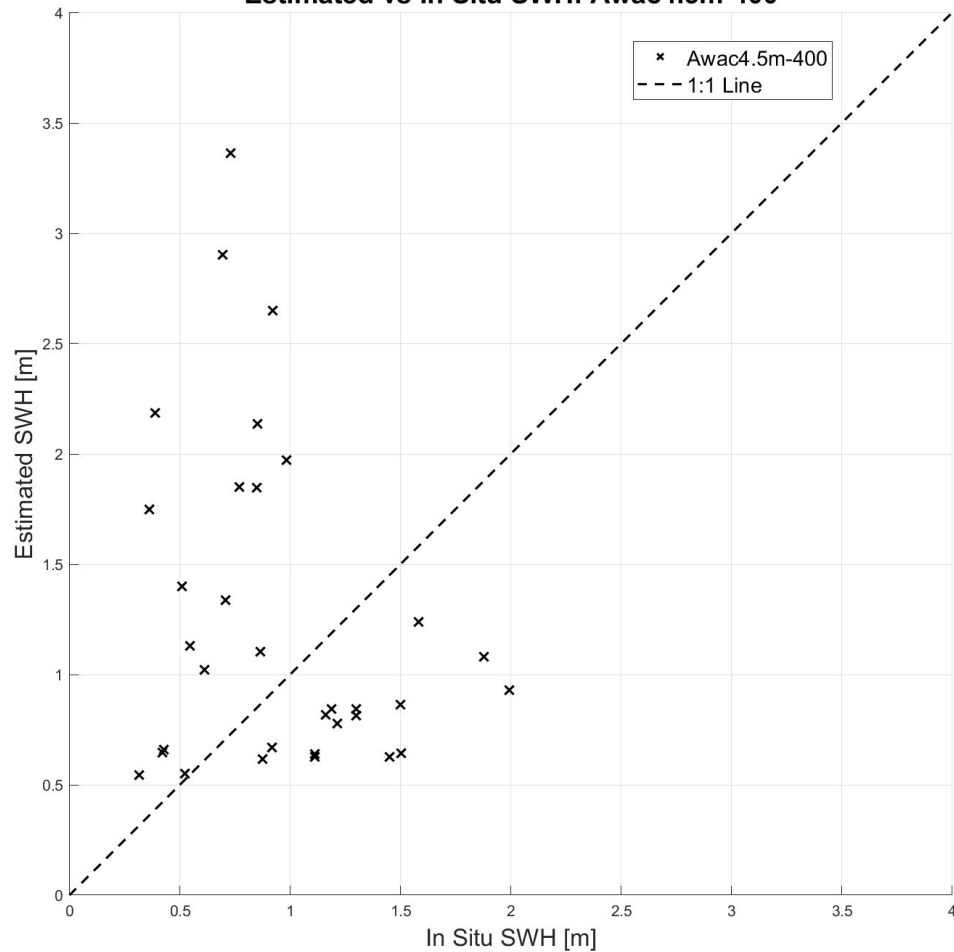
Estimated vs In Situ SWH: 8m_array



Estimated vs In Situ SWH: sig940_600

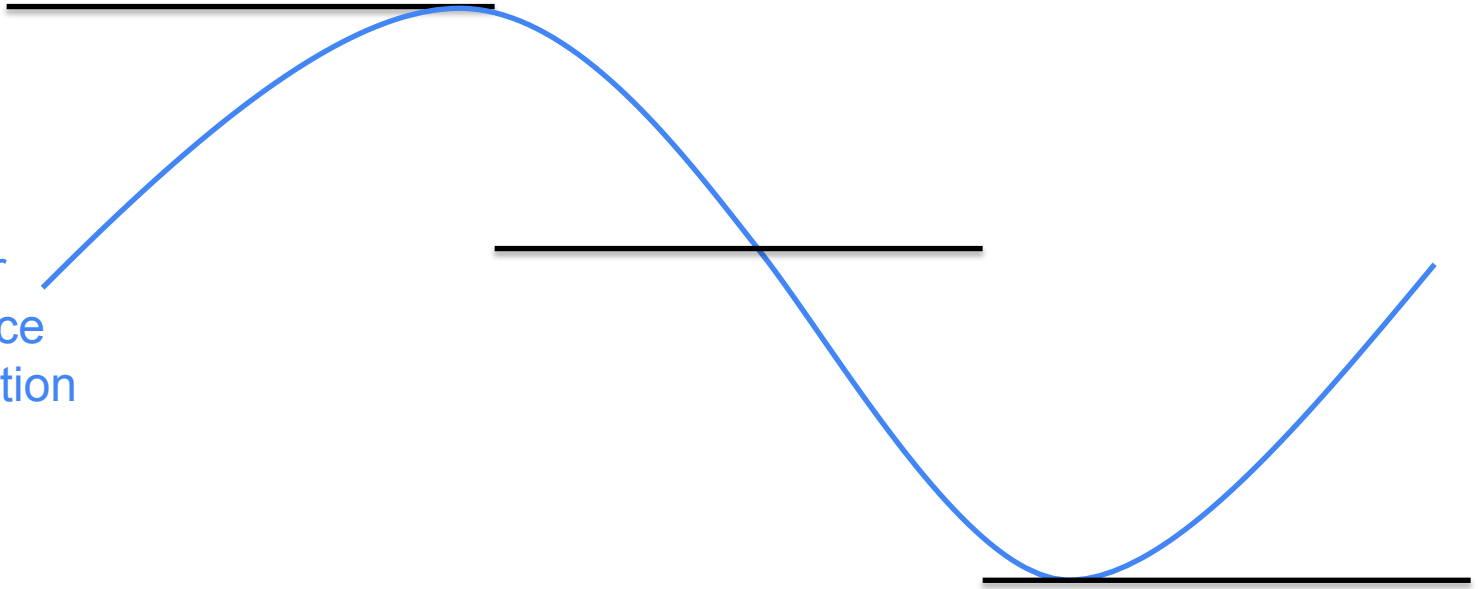


Estimated vs In Situ SWH: Awac4.5m-400

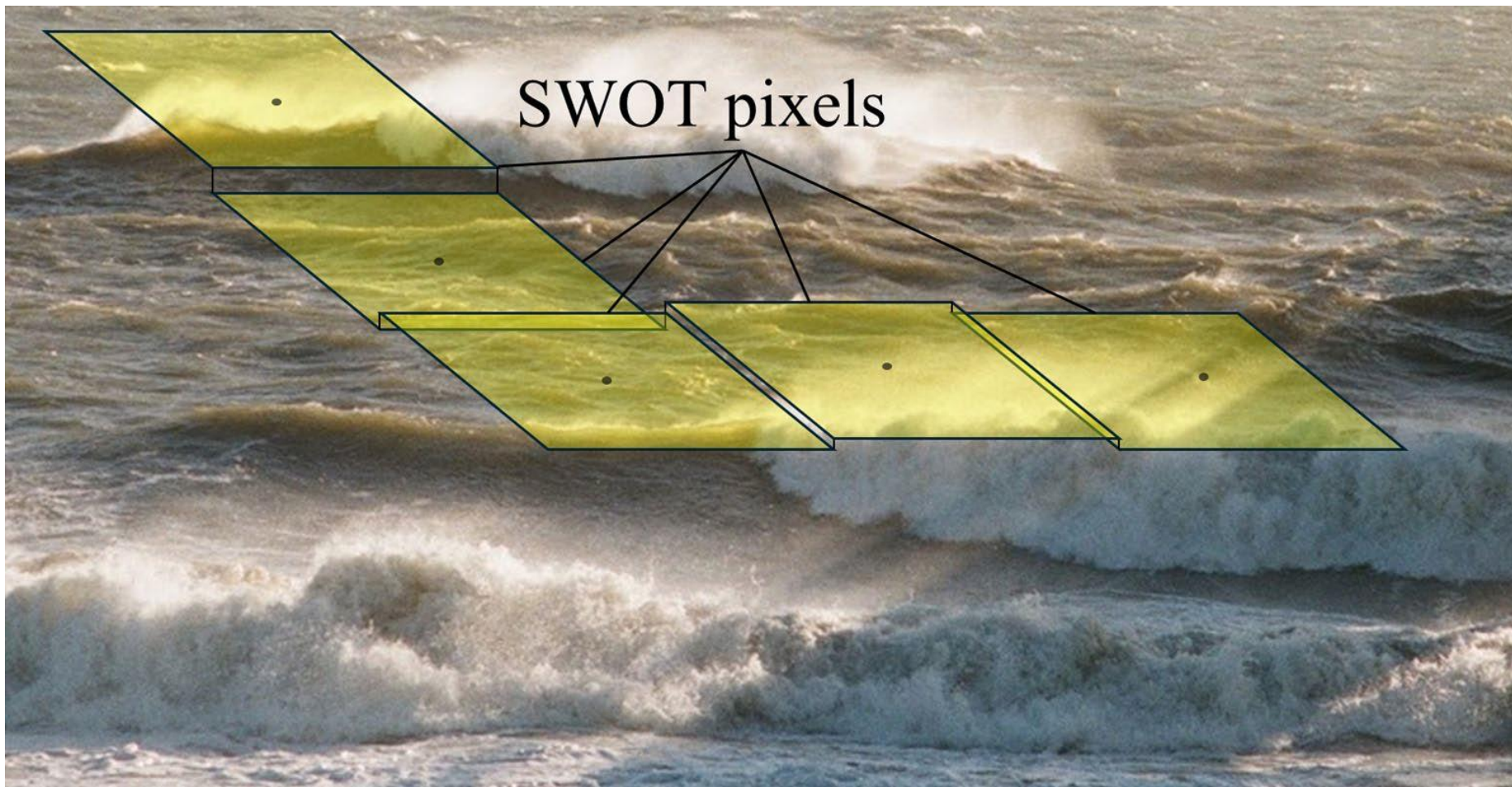


SWOT
Pixels

Water
Surface
Elevation

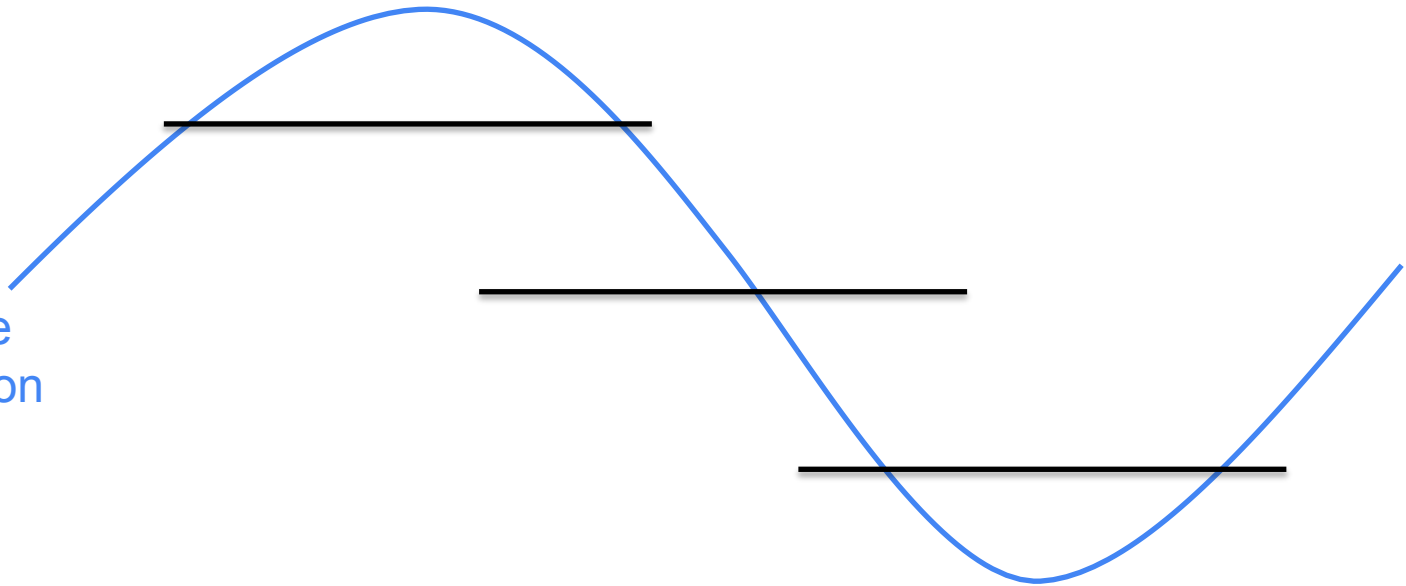


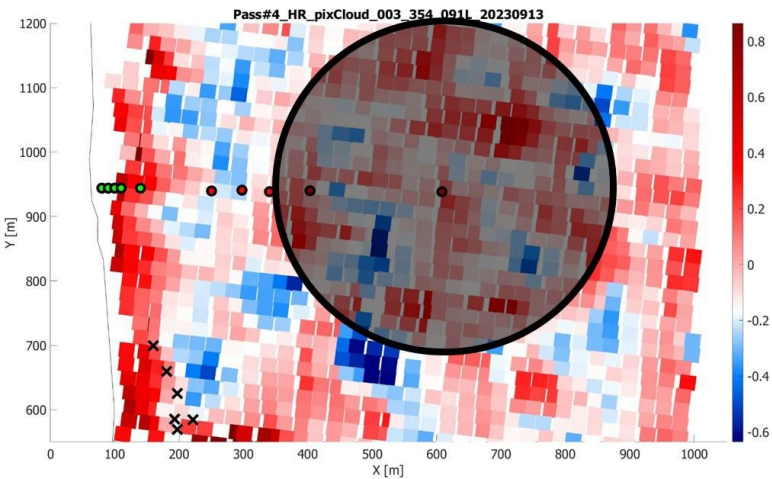
SWOT pixels



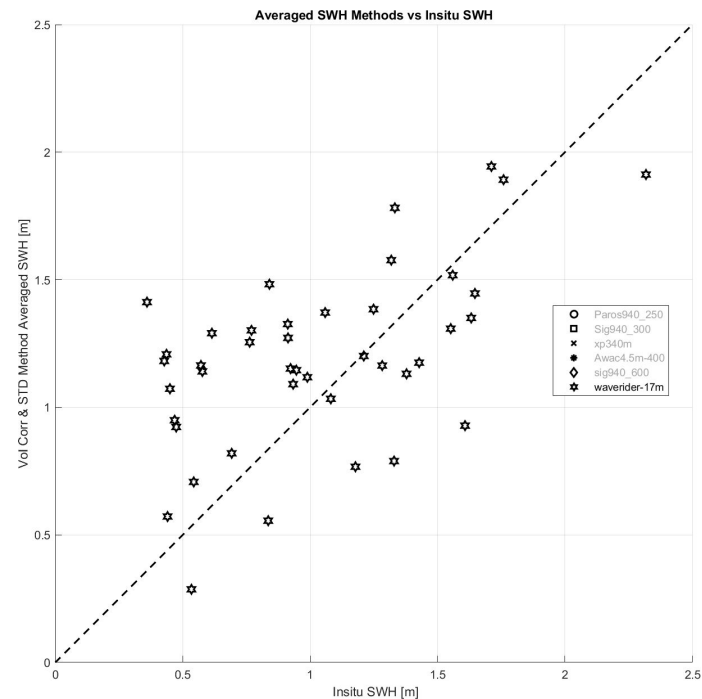
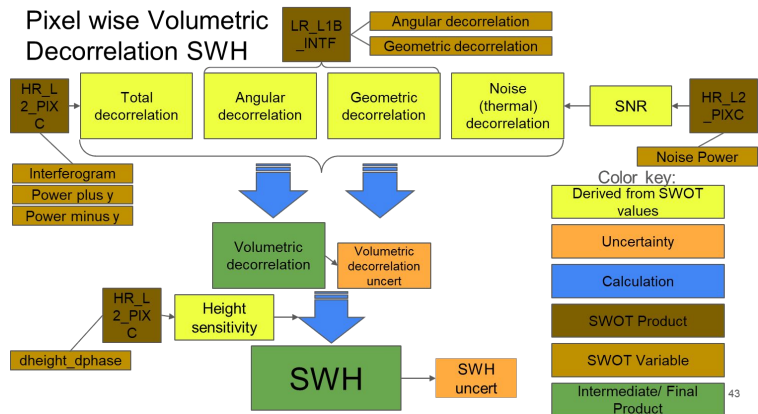
SWOT
Pixels

Water
Surface
Elevation



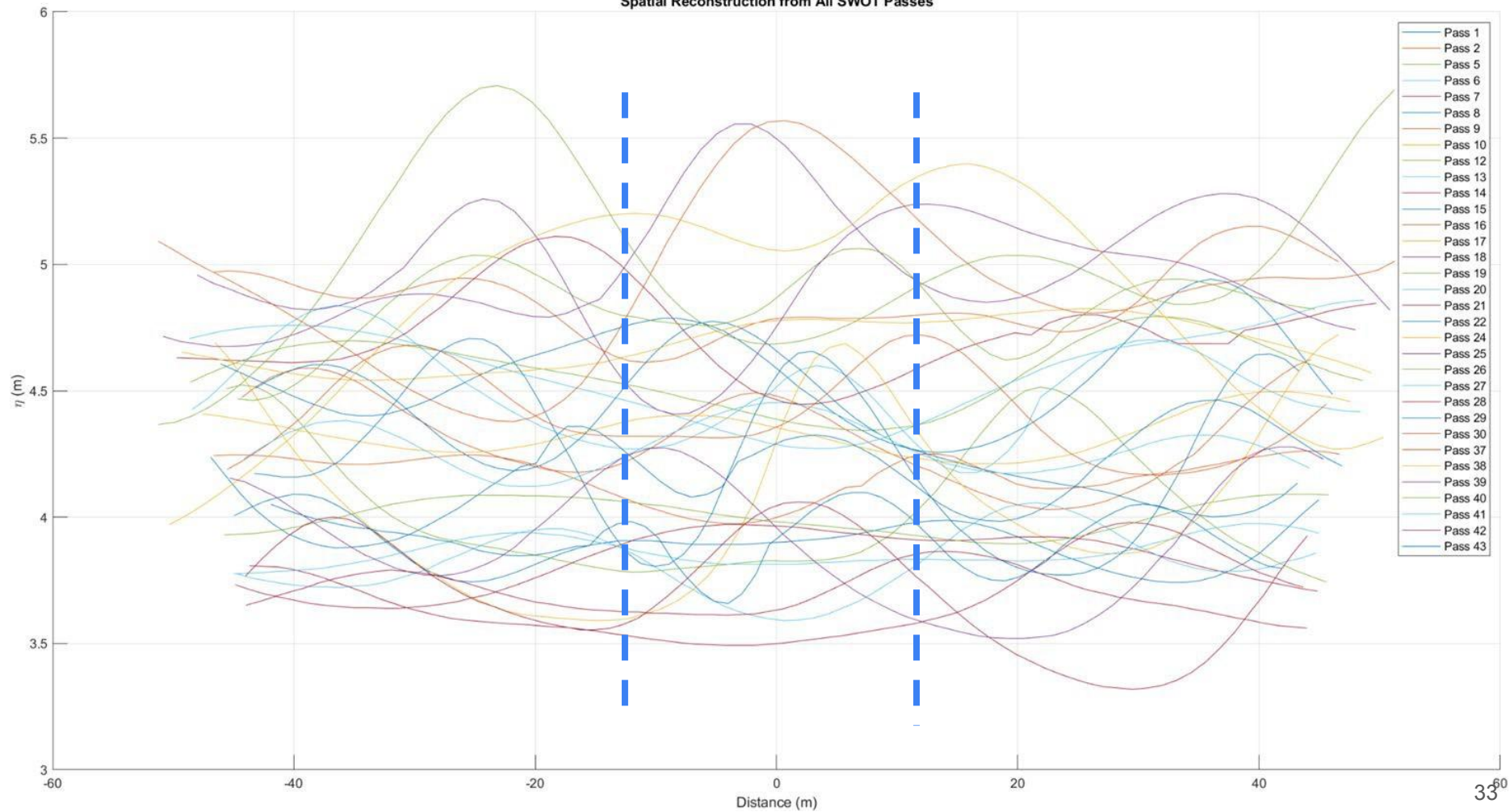


Pixel wise Volumetric Decorrelation SWH

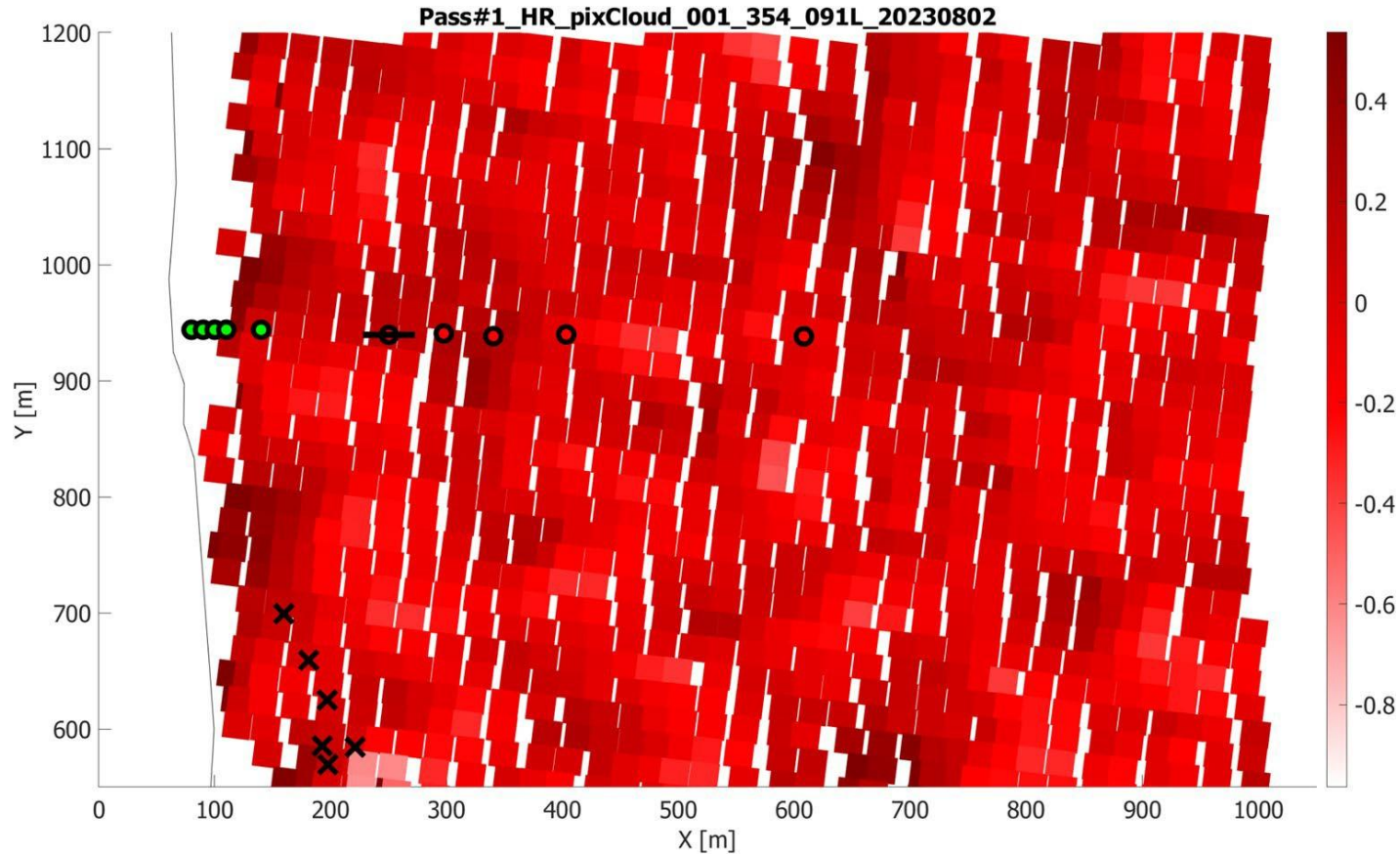


More possibilities with the FRF data

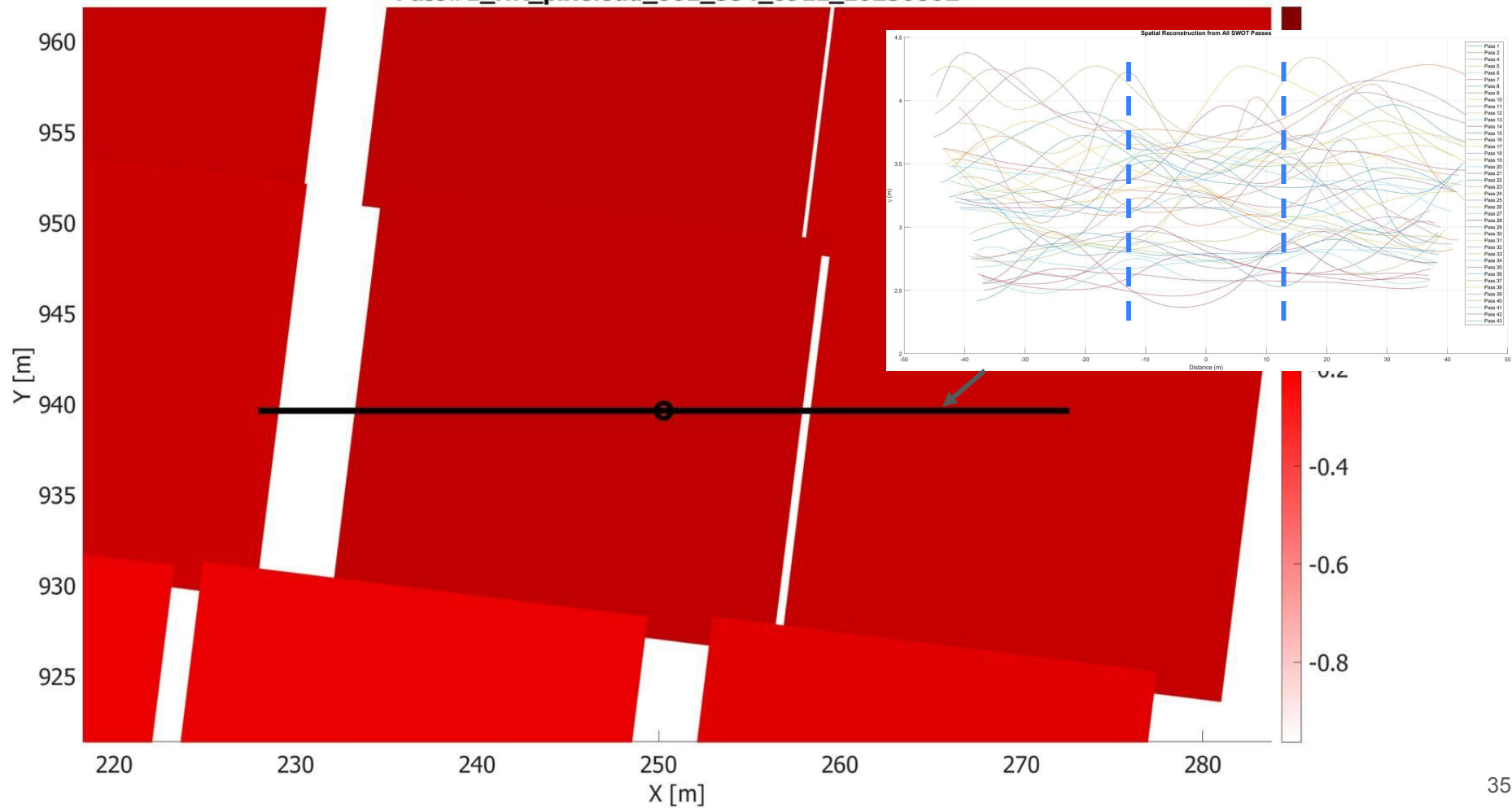
Spatial Reconstruction from All SWOT Passes



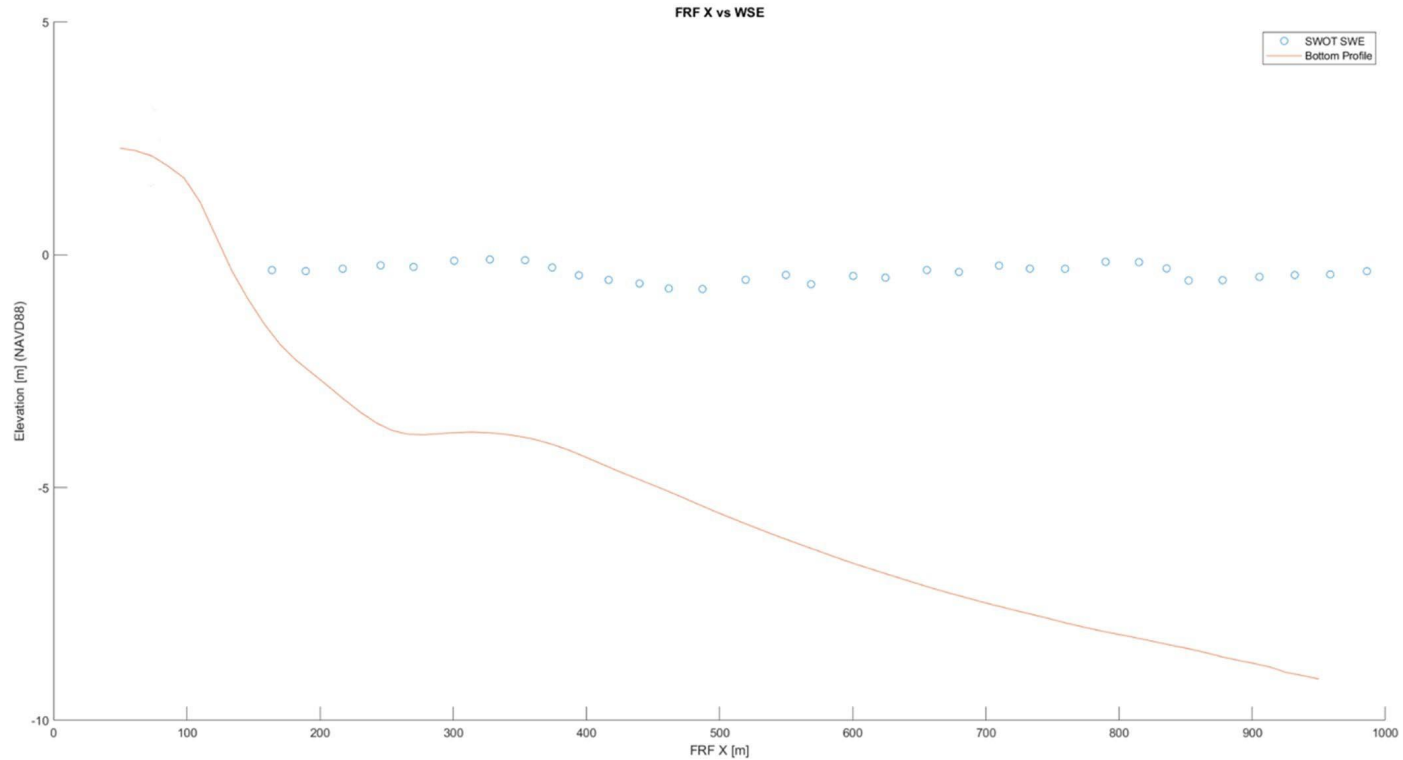
2Hz WSE Data for Surface Reconstruction



Pass#1_HR_pixCloud_001_354_091L_20230802



Bathymetric Inversion Validation



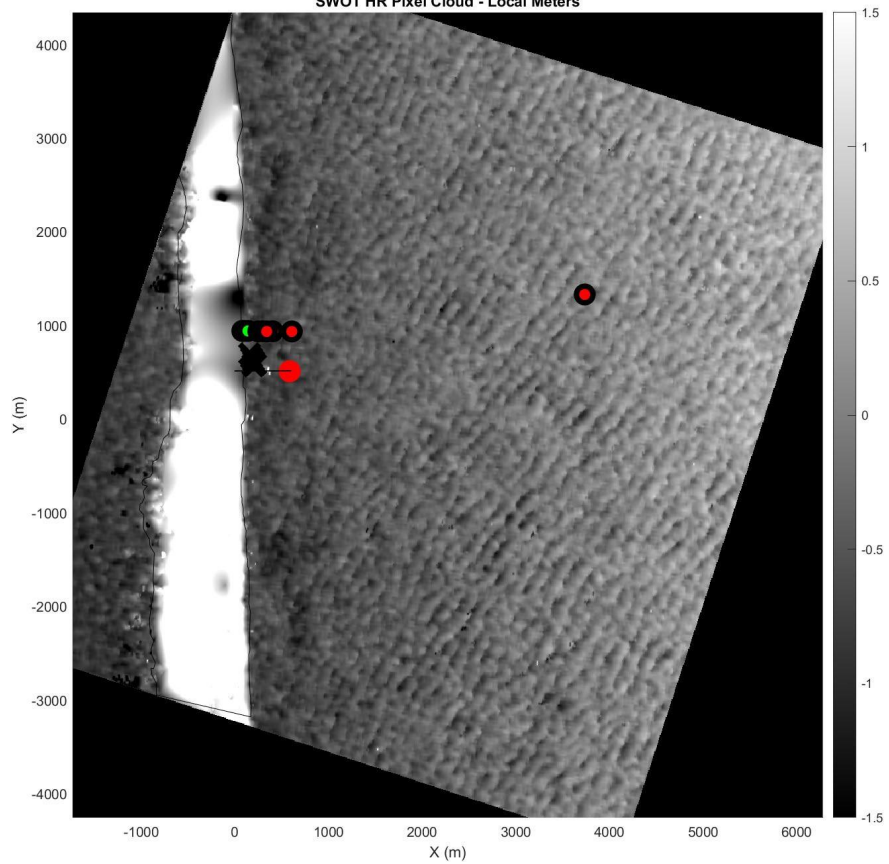
Conclusions

- 2km SWH product reliable until 4km from land (17m water depth), beyond that...?
- Passes parallel to shore (perpendicular to wave field) provide more accurate SWH measurements
- HR Pixel Cloud based SWH measurements show promise for nearshore applications
- US Army Corps of Engineers Field Research Facility valuable in-situ data for coastal SWOT applications

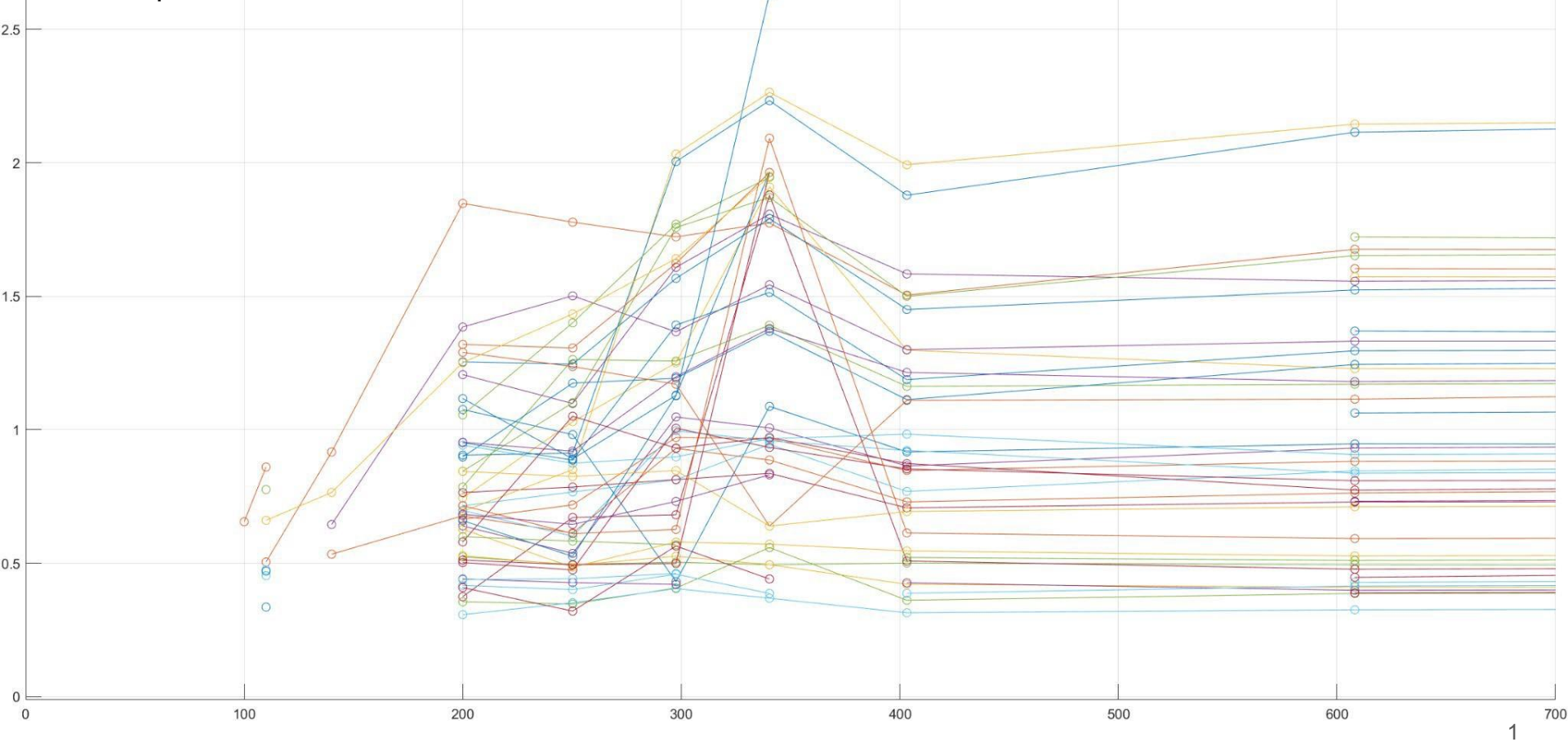


Extra Slides

SWOT HR Pixel Cloud - Local Meters



FRF Wave Heights at each sensor at time of each pass



Pixel wise Volumetric Decorrelation SWH

